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Computer Science Department

COMP433 – Group Assignment phase 1 + 2 +3

<<Project Name: AcademiaHub>>

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PHASE 1:

Introduction

We're excited to share our proposal for a new **Software Portal** designed to revolutionize how Modern Schools operate. Since its inception in 2018, Modern Schools has experienced significant growth, now serving over 500 students with the support of a dedicated team of more than 20 staff members, including teachers, administrators, and support staff. This growth has highlighted inefficiencies in the current methods for managing essential tasks like student registration, attendance, scheduling, and finances. The need for a more advanced, centralized solution has become apparent, particularly in light of the ongoing challenges posed by the COVID-19 pandemic and the growing demand for remote learning options.

We understand that your current portal can be a source of frustration. Many processes are still performed manually, leading to inefficiencies and increasing the risk of errors. Key tasks such as registering students, managing schedules, processing payments, and communicating with parents and teachers are often disjointed, making them harder to manage effectively. Teachers lack a unified platform to share resources and assignments, while parents struggle to stay informed about their children's progress, attendance, and fees. This often necessitates phone calls or in-person visits, creating inconvenience and delays.

To address these challenges, we propose a comprehensive, user-friendly portal that consolidates all critical functions in one place. This portal will streamline administrative processes, enhance operational efficiency, and reduce the likelihood of errors. With a mobile-friendly design, it will automate processes such as student registration, class scheduling, attendance tracking, and grade reporting. Furthermore, it will facilitate seamless communication between teachers, students, and parents, whether classes are conducted in person or online.

From a financial perspective, the portal will simplify payment management by tracking transactions automatically, sending reminders, and applying discounts where applicable. It will also ease payroll processing for the accounting team. Teachers will gain tools to upload materials, assign homework, and monitor student progress, while parents will enjoy secure, real-time access to schedules, grades, assignments, and payment details, enhancing transparency and convenience for all stakeholders.

We recognize the unique challenges faced by Modern Schools and are excited to offer an innovative and practical solution. This portal will optimize school operations, enrich the educational experience for students, and simplify everyday tasks for staff. We are confident that it will position Modern Schools for sustained growth and success in today's dynamic educational landscape.

Proposed Solution

Portal Features Overview Based On Different Departments:

1. Registration Department Features:

Creating accounts and giving roles:

The portal enables registration employees to create individual accounts for all school staff and students and register them in the portal using their essential information, such as their full name, personal photo, and the classes assigned for teachers, and the classes that are assigned for students, and fills additional personal information on those accounts, like their medical information, while specifying who can access them as needed, such as place of residence and medical record and others. Each school staff or student or parent has their own account and can log into it with their special id number and password to ensure their data security and privacy.

The software creates a profile page for each student, parent, and school staff that contains their general personal information including their roles. Also, it gives them the permission to update some of their data as needed.

Student Transfers:

The portal provides functionality for registration employees for transferring students out of the school, the portal ensures that all necessary records(essential information,Grades records , Medical records,Student behavior certificate..) are properly processed, and transferred successfully.

Teacher Transfers:

Registration employees can also handle the operation of transferring teachers out of the school, the portal ensures that all necessary teacher's records(essential information,Contact information,employment history,medical records..) are properly processed, and transferred successfully.

Student Document Management:

allows registration employees to come out with electronic documents reports that students may request, such as transcripts and certificates of grades and behavior. These documents are generated through database queries.

2. Accounting department Features:**Student Payment Management:**

The portal allows the accountant to collect payments from students, ensuring all transactions are recorded in the portal's database and tracked efficiently.

Payment Reminders:

The portal automatically generates and sends payment requests or reminders to parents,if they are late in paying after a certain date, helping reduce overdue payments.

Managing Discounts:

The Software can allocate scholarships for students based on both their grades and financial status. Scholarships are given to the top ten students in the school, in addition to students with average financial status who obtain a grade of very good or higher, in addition to calculating sibling scholarships and exempting teachers' children.

Teacher Salary Management:

The portal manages salary for teachers and other employees, ensuring accurate, timely compensation.

Donations Management:

Accountants can receive donations from parents through the program, allowing parents to transfer funds directly to the accounting department. The donations are recorded in the portal's database, and notifications are automatically sent to the Directorate of Education to ensure transparency.

Salary Deduction Notifications:

Notifications are automatically sent to teachers and employees when the accountant enters a deduction detailing the amount of deductions in cases of unexcused absences, tardiness, or early departures without an excuse.

3. Management Department Features:**Student Reports Monitoring:**

The portal provides the principal with access to various student reports, including grades, attendance records, warnings, and complaints. This allows for efficient monitoring of student performance and behavior.

Teacher Performance Monitoring:

The portal allows the principal to track and review teacher performance, including attendance and any complaints filed. This ensures that teacher performance is closely monitored and managed.

Calculation of class quorum:

The program can calculate the class quorum according to the following equation: $\text{Class quorum for one teacher} = \text{number of classes} + \frac{\text{The number of portions of the material that must be given for each class}}{\text{the number of teachers in that subject}}$. The director enters this information. If the teacher exceeds the quorum allocated to him, which is 25 classes, the program gives A warning that the school needs an additional teacher, and If it results that the teacher received less than the quorum allocated to him, the program gives a warning that the teacher has not completed his quorum, while giving suggestions to the director on how to fill the teacher's quorum in accordance with his specialization.

Creating teachers' schedules:

Our program enables the principal to create the program comfortably, as it detects conflicts and prevents the user from adding the same teacher if he is giving a class at the same time. In addition, it gives alerts if the teacher is placed in more classes than the quorum allocated to him, or if the teachers are distributed unequally in the classes, so some teachers have more breaks than others or have more last classes than others. Our program solves all of these problems. With a smart feature using artificial intelligence, it can distribute teachers to classes according to standards randomly, with the possibility of setting more than one possibility from which the user can choose.

4. Teachers Features:

Teachers have a user-friendly platform to manage their class information and split their students into three sections . The portal allows all teachers in the school to register by creating an account through the register employee.

Schedule Viewing:

The portal provides a clear, user-friendly schedule where teachers can easily view their class times. With a simple daily view, teachers can stay updated on their schedules.

Messaging:

The portal provides a secure messaging feature, allowing teachers to communicate directly with students, parents, and staff within the school portal. Teachers can also send images and documents within messages.

Publishing Learning Materials:

Teachers can upload and share various types of educational materials (PowerPoint, PDF, Word documents, etc.) with students through a dedicated section on the platform for recorded classes and another section for file sharing. This ensures easy access to video lessons and previous exams for students exercising, allowing students to utilize these resources at any time.

Assignment Management:

The portal enables teachers to create and send assignments to students, keeping academic tasks organized and accessible. Teachers can also assign deadlines for each task, and the user-friendly interface ensures that managing assignments is straightforward and efficient.

Grade Publishing:

Teachers can publish student grades directly in the portal, allowing students and parents to view academic grades in real-time.

Teachers office hours:

Our portal sets office hours using ai for the teachers to allocate time for students and parents.

5. Parents Features:**Schedule Access:**

Parents can view their child's schedule, keeping them informed about class timings.

Messaging system:

Messaging feature enables direct communication between parents and teachers, so if parents have any concern about their child or have a question they can message the teachers to solve the issue.

Grade Monitoring:

Parents can easily access their child's grades through the portal, so they can be aware of their child's performance.

Warnings and Alerts:

The portal sends notifications for academic warnings automatically when the child's average goes below the minimum limit, ensuring parents are informed about their child's status.

Payment Management:

Parents can view upcoming payments due in the financial record section in our portal, making it easier to manage tuition and other fees. They can also check their balance sheet to ensure all transactions are accounted for.

Chat Groups:

The program contains chat groups that connect parents whose children are in the same class with each other so they can communicate with each other through texts.

Evaluations:

The portal allows parents to evaluate their children's performance and the extent of their development and their children's progress in terms of academics and creativity.

6. Students Features :**Viewing grades:**

The program contains interfaces that help students view their grades and average division average smoothly.

Messages:

Students can send and receive messages from their teachers, the message portal allows sending pictures and text files.

Assignment management:

The platform allows students to submit assignments, research papers, and visual presentations, including tracking assignments, deadlines, and submission statuses.

Evaluations:

Our program allows students to evaluate the teachers' performance, the way they teach, and the extent of their benefit at the end of each semester.

View schedules:

Each student can see his class schedule in addition to the class group.

7 . Secretary Features:**Managing books and Uniforms:**

Our program facilitates the work of the secretary by facilitating the process of receiving and delivering books and school uniforms by collecting their price and calculating their preparation automatically. Through the program, the secretary can also record all in-kind donations and gifts to the school, count their number and purchasing value, and send notifications to the Director of Education about them.

Scheduling:

The secretary can schedule and hold direct or remote meetings for the principal with teachers or with the parents' council at appropriate times.

8 . Cleaning Employees Features:**Identifying cleaning areas:**

The program identifies for cleaning workers the areas assigned for each person clean, with the ability to display a drawing of the map of school. The map will display the parts assigned for the worker to clean coloured red which regrests to be "unfinished", and upon completion of them, the cleaning employee can change the color to green which refers to "finished".

Task List:

The portal features a task list to help workers stay organized. After completing a task, they can cross it off the list, with the time of completion . This tracking provides a record that can be used to investigate incidents like theft.

9. Additional features:**Dealing with disabled students:**

Our portal makes it possible to deal with all categories of students, including those who suffer from learning difficulties, by presenting simplified curricula that tend towards the visual nature, and it allows the conversion of texts into the spoken and audio portal for those with blindness and the conversion of spoken text into readable for those with deafness.

Helping students choosing a field:

For high school students ((tenth grade)) approaching the eleventh grade, our program portal analyzes the students' grades and based on the results, the portal suggests which branch to advise the student to pursue (scientific, literary,..) while displaying an approximate probability of his chances of success in each constructive branch. On the mark sheet.

Interactive educational games:

The portal contains interactive educational games, and the “Test Yourself” feature, electronic exams with a self-corrector, to help students prepare for their exams and identify the type of questions.

Help determine future job:

For high school students (Tawjihi), our portal enables them to determine the appropriate university major for them based on a short test that they take through the application, in addition to an analysis of their transcripts for various subjects (physics, geography, chemistry...) so that the result is very realistic.

10. The Non-functional requirements:

We're excited to introduce a high-performance, secure, and scalable school management portal specifically designed for your school's long-term success. Our solution goes beyond basic features, ensuring that core functionalities work smoothly while focusing on **speed, reliability**, and software's **usability**.

Our portal will be designed to handle peak loads efficiently, especially during busy periods like new terms beginning and grade releases. We took the number of users you provided in consideration so, our portal will deliver quick page loads and fast processing, even with hundreds of users using the portal at the same time. We also care about portal's scalability. As your school grows, the portal scales effortlessly, containing more students, teachers, and new services such as extracurricular programs or digital libraries.

We also prioritize the **security** of students, parents and school staff data through robust role-based access control. We also care about the usability for this software. We are aware of the fact that we are dealing with users of various ages, so we put huge effort in creating a friendly-user clear interface and if you give us the honor to work on this project, we are willing to develop a simple guidance portal for new users in the future.

And since we are very welcoming to your feedback our portal will be well structured so that it's **easy to update and modify**. **And the portal will be working 24/7**, allowing parents and students to access assignments and resources anytime remotely.

Software development process

Incremental development process particularly Agile will be used in this project. We are going to apply scrum in this project because the requirements are incomplete and we need to communicate with the customer continuously at the start and end of the sprints. We decided on this working method since the approach is incremental and adaptable and promotes ongoing development. Agile enables prototyping and gives schools the chance to test new features and requirements, providing continuous feedback that helps us understand the portal and continue developing it to meet the needs of users.

Our software is based on a five phase cycle per sprint as follows:

First gathering requirements from customers through a meeting that will be held at the beginning of each sprint, features will be written in a detailed

description based on the customer's needs. Both UI and UX will be present to design sketches and provide prototypes to help the customer understand the portal clearly and greatly. Then Both UI and UX will develop the interfaces based on the sketches

drawings agreed upon in the meeting .After, developers will implement the code for the designs. Next, Testing; Once the code is built, it will be tested and all possible cases will be covered. Finally, After completing the work and making sure that the tasks are perfectly done by the teams, a meeting will be held on the last day of the sprint to present the finished work to the client and take feedback, changes are discussed and other requirements are listed to be delivered and implemented in the next sprint.

These steps will be repeated until the project is completed.

First release :

In this release, the most important and most complex requirements will be delivered. The feature of creating accounts and assigning roles to the registration department will be delivered. In the Accounting section, the student payment management feature will be developed due to its importance to both the student and the school. Moving to the management section, the manager will have the authority to create schedules for each teacher and another feature is calculating and determining the legal quorum for the class. In another section, the teachers section, several features will be built: the teacher's ability to access his schedule and his ability to send messages to the relevant party. Since the teacher directly affects the student, at this stage he will be able to send assignments and show grades to the student. In the students section, he will start submitting his assignments and sending messages to teachers. Of course, he will be able to access his schedule and class times. Since parents have a special section for them, one of its requirements is his ability to access his son's schedule and send messages. He can also see his grades. Given the importance of the secretary's section at the beginning of the semester, the feature of distributing books and school uniforms will be created in this version due to its reliance on the Ministry of Education.

This release is expected to take not less than six months

Second release :

In the second release, The teacher is capable of distributing educational resources to his students, as depicted earlier, while the student will be able to see his grades. At this stage, parents will receive warnings automatically, as in the description, and a payment reminder notification. This release will enable the accounting department to send a reminder notification to pay school fees. The teacher salary management feature will also be activated. It will be able to send a salary deduction notification to teachers. In the registration section, you will be able to use the student and teacher transfer feature for the purposes of transfer and transportation outside the school, and it will be able to prepare and manage student documents through the student document management feature.

The completion of this release is projected to be three months after the first release is delivered.

Third release :

At this time we will have reached the third release in which the scheduling feature will be applied and implemented in the secretary section as discussed before, and thus this section will be completed finally. In the management area, the manager will have the ability to create comprehensive reports evaluating both student and teacher performance. These reports will be based on the various factors discussed previously. As for the financial section, it will be able to manage donations and discounts on student fees. With this, we have finalized this section and this release.

The third release is scheduled for two months after the initial releases.

Fourth release :

The last release will be dedicated to finishing the project and completing its requirements, which are creating a special section for cleaners and establishing special features for them, which are the identifying cleaning areas feature and the task list feature. As for the teachers' section, it will end with creating the office hours that each teacher determines. As for the students area, it will be programmed with a feature for disabled students called dealing with disabled students, as well as an important feature, which is helping students choose a field, and also another requirement for entertainment, which is an interactive educational game. Also, students will have a feature that helps them choose the appropriate branch for them in the Tawjihi stage based on the results of a small test they submit and also based on their grades at school. Thus, the students' section has been completely completed (all of them have been detailed previously). In the last section, which is the parents' section, Parents can set up private groups, providing a dedicated space for them to connect and interact.

This release will be finished in approximately 1.5 months.

We expect that the time required for this project will be about a year from the start of work on it. A confirmed date will be given according to the team's speed of work, which will appear in the first sprints. We also assure you that development is certain in this project in the coming years, hoping that you like our ideas.

PHASE 2:

Task 2.1: [Group Task]

User Requirement #1:

1.User Registration: The portal shall allow registration employees to create individual accounts for all school staff and students and give each user specific permissions based on their role, and after registering them successfully it should allow the users to login into their accounts.

System Requirement #1:

1.1: The portal shall allow the registration employees to create accounts for each user, filling a form with user essential information which are: (user ID, full name, personal picture, email, password, assigned classes). And additional personal information like medical information which are: (Allergies, Chronic diseases, Previous vaccinations, Medications, Previous medical operations).

1.2: After filling the information form, the registration employee assigns a role for the user which the roles are (student, teacher, accountant, manager, parent, secretary, cleaning employee).

1.3: The user shall be able to login into their account using user ID and password.

1.4: The portal should ideally respond to login requests within 0.1 to 1 second, if immediate response is not achievable, the response time shall not take more than 2 to 3 seconds.

1.5: In case the user forgets their password the portal will offer an option for contacting the registration department to assign a new password, new password will be sent to the user's registered email address or delivered via SMS to their registered phone number.

User Requirement #2:

2. Student payment management: The portal system shall allow accountants to acquire student funds to ensure that the accurate financial documentation is in the portal data archive.

System Requirement #2:

2.1: At the beginning of each semester, each student will check out the enrollment fee that he must pay for the current semester according to the student's academic level.

2.2: The portal shall monitor the tuition fee payment status of all students and, at the end of the two-week payment period, identify students who have not completed their payment. Once identified, the portal shall generate a notification message and deliver it through the portal to remind those students to pay their tuition fees.

2.3: The message shall contain date and time of dispatch, the school logo, and name of the school headmaster/headmistress, name of the student, School ID number, his/her academic level, and the tuition fee.

2.4: The portal shall prepare a detailed and complete report for students who have settled payment for their school fees by the end of the first two weeks of the semester.

2.5: The report shall include the Full name of the student, student ID, student's personal ID number, father's surname, mother's surname, two phone numbers, academic level,

amount of the installment paid, payment date, mode of payment (cash/check), number of checks if payment method is by checks, and a payment acknowledgment receipt.

2.6: The payment acknowledgement receipt shall possess the following data: (date and time of dispatch, the school logo, and name of the school headmaster/headmistress, name of the student, school ID number, his/her academic level, date and amount of tuition disbursed).

2.7: The system shall store the prepared reports securely in the data archive. Each payment shall update the outstanding balance, with the payment details, including the date of payment, logged in a separate transaction record.

User Requirement #3:

3. Users messages: The portal provides a secure messaging feature, allowing teachers to communicate directly with students, parents and staff within the school portal .And also allowing students and their parents to communicate with the Teachers .They can send images and documents within messages.

System Requirement #3:

3.1: Teachers can communicate with their students and their parents and send different types of text messages, files or videos through our user-friendly interfaces . And the same for the students and their parents.

3.2: There is an archive of messages sent from previous days, months, and even years. They are archived during a specific time specified by the sender, or they can be left for the program's default time to archive the message a week after the end of each academic year. The user can then return to any message he needs at any time before the end of its default time.

3.3: There is an option to search for a message by date and time, sender name or id , or recipient name or id. Messages can also be sorted in ascending or descending order based on the date sent or ended, sender or recipient id, or alphabetically based on sender or recipient name.

3.4: Through the messaging feature between parents and teachers, parents can stay fully informed about their children's level and behavior

3.5: The student can send messages easily and instantly, specifying who receives them by either entering the student's id, the designated teacher, or the full name. He can also send a group message to all students in his class, with the option to exclude or not exclude the class teacher from that message.

3.6: The portal enables the teacher to communicate with school employees, which helps teachers of the same subject to agree with the common group

3.7: Send notification when receiving any message

User Requirement #4:

4. Teacher schedule viewing: The software portal shall allow teachers to view their schedules online, keeping them informed about their class times and any additional commitments.

System Requirement #4:

4.1: The portal system shall let teachers see their daily schedules, with class timings displayed in an easy-to-read format.

4.2: The portal system shall display the schedule in a way that helps teachers quickly understand their daily commitments, including classes, meetings, and training.

4.3: The portal system shall allow seamless navigation between days and weeks, so teachers can easily check their upcoming schedules.

4.4: The portal system shall update any changes to the schedule in real-time, ensuring teachers always have the latest and most accurate information.

4.5: The portal system should include a search feature to help teachers quickly access and view schedules across all years.

User Requirement #5:

5. Creating teacher's schedules: The portal system shall allow the principal to create and manage teacher schedules efficiently. It shall automatically detect and prevent scheduling conflicts, such as double bookings of teachers for multiple classes at the same time. The system shall provide a mechanism to verify that schedules are conflict-free and adhere to predefined constraints.

System Requirement #5:

5.1: The portal system shall validate the selection of a teacher for scheduling by requiring the teacher's User ID and full name to ensure accurate identification.

5.2: The portal system shall detect scheduling conflicts when a teacher is assigned to multiple classes at overlapping times, preventing the addition of such conflicting schedules.

5.3:The portal system should provide alerts if a teacher is assigned to more classes than their allocated quota, ensuring a balanced distribution of teaching responsibilities.

5.4:The portal system shall monitor the distribution of teaching assignments to ensure that no teacher has significantly more breaks or late classes than others, promoting equity among staff.

5.5: An artificial intelligence option shall be available to automatically assign teachers to classes based on predefined criteria. The principal shall have the option to enable this AI-driven scheduling feature or opt to create the schedule manually. Multiple scheduling configurations shall be provided for the principal to select from.

5.6: The portal system should ensure that scheduling requests are processed and displayed to the user within three seconds to maintain workflow efficiency.

5.7: The portal system shall notify teachers of any schedule changes in real-time, ensuring prompt communication. Notifications shall be delivered via email, SMS, or in-app messages based on user preferences.

5.8: The portal system shall maintain a log of all notifications sent, allowing users to review past changes and alerts.

5.9: The portal system shall provide functionality to analyze teacher workload and class distribution. It shall generate detailed reports identifying imbalances in teaching loads, enabling review and corrective action.

5.10: The portal system should allow reports to be exported in different formats, such as PDF and Excel, for analysis or sharing with stakeholders.

User Requirement #6:

6. Students viewing schedules: The portal enables the student to obtain his/her weekly schedule and the effects that occur to it, in a smooth and easy-to-use way.

System Requirement #6:

6.1: The portal enables the student to see his weekly class schedule, which includes the days of the week and class times, in addition to the names of the subject teachers in a dedicated column next to each subject.

6.2: The option to view the weekly schedule allows the student to see the schedule in a clear and easy-to-understand manner that matches the current time, as the current day of the week is shaded to help the student prepare his schedule.

6.3: The portal also marks the days that contain exam dates for the student to alert him. A mark is placed on the classes that contain an exam, and when you click on it, the specific subject from the curriculum for the exam appears (from page to page..)

6.4: The portal shows any changes that may occur to the schedule, such as changing classes to work classes or changing the subject teacher, with a notification being sent to the users.

6.5: On the class schedule, our program places a mark next to the class of the next day, in which the location of that class may change, for example, giving the science subject in the laboratory instead of the classroom on that day, designated by the teacher for the students, and sports classes and places for work classes (library, playground, classroom). with Sending notifications to students to alert them

6.6: The portal sets the holidays of the week and shades them with a different color to remind and send notifications to students, in addition to the days that contain important dates such as receiving certificates and others.

User Requirement #7:

7. Students and parents viewing grades: The software enables the student and parents to view the marks and their accessories in a very easy way that is suitable for users.

System Requirement #7:

7.1: The portal contains a special interface that enables the student and parents to view the grades that the teacher publishes. They can also see the percentage of the exam or assignment from the full mark and the percentage of the average for the entire semester.

7.2: The program displays the marks divided into daily exams, monthly exams, and final exams for each semester.

7.3: The portal also allows the student and their parents to see the grade in a single subject after collecting the marks of its exams (excellent, very good, acceptable, etc.) next to that subject. With an emoji to give a visual touch.

7.4: Our software also displays the subject teacher's note in a special box so that the student can take the note into consideration and improve his/her situation.

7.5: The portal also allows the student and their parents to see their rank in the class and at the school level through a dedicated window.

7.6: The software contains a complete record of the student's grades in previous semesters and years.

7.7: Receive notifications when new marks are released.

7.8: Our software is connected to an artificial intelligence portal that gives the student and their parents information about the student level in the subjects compared to his previous grades, from an evaluation of his/her performance and some instructions in the event of a decline in his/her grades about ways to improve and the extent of the decline,

and the same thing if the student's academic level increases, the program gives an encouraging phrase and a comparison between the current and previous grades, with all of this explained in graphs.

7.9: Parents can see their children's grades based on behavior, cleanliness, commitment and attendance.

User Requirement #8:

8. Publishing grades: Our portal enables the teacher to publish students' grades easily so that students and parents can view them.

System Requirement #8:

8.1: With our software, teachers can enter students' grades and publish them instantly and seamlessly through our user-friendly interfaces.

8.2: Our portal contains fields for the basic marks that the teacher must enter from daily, monthly and final exams, as well as fields for marks for behavior, activities and assignments.

8.3: The mark box contains a special icon that enables the teacher to collect attendance and absence and collect marks on them. The portal sends an immediate message to the parents if their child is noticed to be absent from any class, and if the student exceeds 15 days of absence from each class without an excuse, he is automatically considered to have failed.

8.4: The portal makes it easy for the teacher to calculate the semester average, as this is done automatically by calculating the sum of marks divided by the number of students in the class, and it also allows the teacher to enter the percentage of the specified mark from the final total manually.

8.5: Our portal calculates the failure and success rate automatically using artificial intelligence, showing graphs of the rates and giving suggestions to the teacher if there's a need for a curve up or other solutions.

8.6: The portal arranges the students in the class in ascending or descending order based on their marks so that the teacher can know the top students in the class.

8.7: There is a search option that allows the teacher to search for a student by full name, part of the name, or id.

8.8: At the beginning of each semester, the portal allows the teacher to create his own division of marks for each semester and to divide the marks based on the appropriate criteria that will apply throughout the semester, with the possibility of modification at any time.

8.9: The teacher must obtain two marks from the short tests. It is possible for the teacher to give more than one test daily, so our portal automatically calculates the two highest marks for the student, with the teacher always having the ability to modify as he sees fit. The same applies to assignments and project evaluations according to the teacher's division of marks.

8.10: Our software sends notifications when the teacher posts a new mark for the students.

User Requirement #9:

9. Teacher salary Management: The portal system shall calculate and process salaries for teachers and other employees by the 7th of each month. It shall account for deductions , generate payslips, and initiate direct bank transfers.

System Requirement #9:

9.1: The designated accountant shall pay the teachers' salaries through an option available in the portal system.

9.2: The teacher's salary is determined according to the following: seniority and academic degree. The salary increases in direct proportion to the increase in these two factors.

9.3: The basic salary shall be 2,000 shekels for all teachers, with a salary increase every five years of 100 shekels for everyone. Regarding individuals possessing a master's degree, it will be 300 shekels, In contrast to those with a doctorate, whose increase reaches 600 shekels per month for both of them, Plus an additional allowance to offset the high cost of living that reaches 150 shekels per month. The salary can reach 5,000 shekels. Once this amount attains the level of the salary, salary advancements will terminate.

9.4: A salary penalty is applied for unauthorized absences for teachers , i.e. in the event that there is no medical justification, and a daily deduction of 100 shekels is imposed. Teachers are entitled to a maximum of 3 consecutive days of paid sick leave per illness. For absences exceeding 3 days, teachers must present a medical certificate from the Ministry of Health for review by a medical committee. If the medical committee approves the leave, it will be considered paid sick leave. However, if the leave is not approved, a deduction of one day's wage will be applied, calculated as the teacher's salary divided by 22 hours. It is also possible to deduct from the teacher's salary for continuous delay, and the deduction will be made when collecting the hours of his full day's work.

9.5: The portal system shall determine the amount of the monthly salary that teachers will receive according to the increases and deductions mentioned above. The salary is paid to teachers after the seventh day of each month and may be delayed until the tenth day of the month, via the individual private bank accounts of each teacher.

9.6: After the salary reaches the bank account, the following salary information Will show up to the teacher: Base salary, salary raise, salary decrease, adjusted salary, and salary payout date, and through an option, the teacher will be able to know the reason for the increase and deduction of the salary according to the previous matters.

Task 2.3: [Group Task]

Effort & Cost Estimation:

To calculate the Effort and Cost Estimation we will go through some steps:

1. **Determine main features:** We have **9** main features.
2. **Determine the complexity:** We give each requirement function points, these function points determine the complexity of the requirement , we will use a rate from 1 to 5 (1 means least complexity, 5 means most complexity).

Below, a table describing each requirement complexity :

User Requirement	Complexity
UR1. User Registration	2
UR2. Student Payment Management	4
UR3. Users Messages	4
UR4. Teacher schedule viewing	3
UR5. Creating Teacher Schedule	5
UR6. Student Viewing Schedule	2
UR7. Student And Parents Showing Grades	3
UR8. Publishing Grades	3
UR9. Teacher Salary Management	5

3. **Calculating the summation of function points:** Calculating the total function points of all the requirements.
4. **Determining the conversion factor:** The conversion factor represents the amount of work hours needed to complete one function point, we determined the conversion factor as **12 hours**.
5. **Calculate the effort:** Calculate the effort for each user requirement

$$\text{Effort} = 12 * \text{function points of requirement} = 12 * 31 = \mathbf{372 \text{ hours}}$$

6. **Determine working hours:** Determine how many hours **one developer** works a day.

below, a table that shows the **total function points, total effort, number of days required for one developer to finish one the total of user requirements:**

User requirement	Function Point	Effort = Total function point * Conversion Factor (12 hours)	Number Of Days required for one developer
UR1	2	24	4
UR2	4	48	8
UR3	4	48	8
UR4	3	36	6
UR5	5	60	10
UR6	2	24	4
UR7	3	36	6
UR8	3	36	6
UR9	5	60	10
Total	31	372	62

- To complete this project, one developer will take **3 months** of work, so if we distribute the work to **three developers** , we will need **1 month** to finish the whole project.
- To calculate the cost of a programmer on a monthly basis who will work **100%**:

- The senior developer earns a salary of **\$3000**.
- The first mid-level developer earns **\$1100**, and the second mid-level developer earns **\$1200**.
- Together, their salaries amount to **\$5300**. When combined with operating expenses of **\$700**, the total cost reaches **\$6000**.
- The **min** profit margin for this project is **30%**, which translates to **\$7,800**. The **max** profit margin is **50%**, or **\$9,000**.

PHASE 3:

Task 3.1: [Individual Task]

1- Student payment management Use Case (Toqa Abdeen) :

Baseline supposition:

- The first two weeks of the semester have ended, i.e. the end of the tuition payment period. The accountant logged into the system to access the school's data archive, which shows the basic information for all students, which is the following: student's personal ID, student ID, father's surname, mother's surname, two phone numbers, academic level, and tuition payment status (paid/unpaid).

Normal:

- Regular mode: In standard operating procedure, the accountant identifies students by their assigned numbers and subsequently selects the corresponding tuition fee from a designated list to finalize the entry of financial data.
- The system then guides the accountant through a series of steps, requesting input of financial information in various fields, including the amount of the installment paid, the

date of payment, the method of payment (cash/check), the number of checks if the method of payment is by check, and a payment acknowledgment receipt, by selecting the "Save" option, the accountant will finalize the student's financial information. The system will then immediately update the student's tuition payment status to "paid" in the data archive.

Alternative:

- While still logged in, the accountant investigated the accounts of students who had missed the payment deadline, but he found the following conditions in the data of some students: economic distress, medical emergency, administrative error, hold-up in external funding, request for a payment plan. Therefore, the accountant and the financial department decided to either grant an additional grace period or reject the application based on their review as follows:
 - ❖ The first student was experiencing financial distress. The accountant noticed this when the student submitted a formal request to extend his tuition payment period. Additionally, he presented a letter of explanation outlining his financial situation, accompanied by supporting income documentation. Upon review by the accountant and finance team of his request to determine if his difficulties qualified him for a grace period, they granted the student a one-week extension and the accountant sent a letter confirming the approval of his request and the extended deadline for payment.
 - ❖ An accountant, after completing his investigation, identified a student who had failed to pay tuition fees on time due to a medical emergency. The student provided valid medical documentation explaining the illness that caused the delay in payment. The accountant and his team carefully reviewed the submitted documents and, considering the medical condition to be non-serious, denied his request to extend the payment deadline for an additional three days. The accountant then sent an official letter to the student explaining that the request was denied and that he was given an additional three days instead of two weeks.

- ❖ The accountant delved deeply into his research and found that some students had fallen into various administrative errors, such as delays in confirming eligibility for fee exemptions or study grants. After discovering this error, the accountant approved the student's tuition waivers. The accountant accessed the data archive, located the student's record, and modified the payment status to "Paid" by clicking the "Save" button. An automated message was then sent to the student indicating that their tuition fees were due.

- ❖ The accountant found another student who had submitted a registration application after the deadline, but due to administrative delay, his application was not processed on time. After resolving this error, his late registration application was approved, the grace period for paying tuition fees was extended, and a notification was sent to the student confirming the approval of his late registration, the new payment date, and the amount for the tuition fees.

- ❖ The accountant identified another student affected by administrative mistakes, which is the delay in external funding, where the student relies on an external party to pay his tuition fees (scholarship, sponsor, government grant), but due to reasons beyond the student's control (administrative delay from the funding source), the disbursement of these funds was delayed. The accountant understood the student's situation because the student submitted an official request through the portal in which he informed the school of his situation in addition to supporting documents that proved the error. The student submitted a sponsor's letter in which he explained the delay in funding and the expected disbursement date. The accountant completed his work with the financial department by verifying the validity of these documents, and the verification was successful. As a result, the student was granted an additional payment period of one week due to the expected disbursement date from the sponsor's letter. Accordingly, a notification was sent to the student explaining the approval of his request and informing him of the new payment date.

- ❖ A student, identified by the accountant due to administrative errors, submitted a request through the portal for a payment plan because of financial difficulties and an inability to pay the full tuition fees on time. The accountant accessed the form completed by the student, which included responses to the following questions:

- **Reason for the request:** The student selected "financial distress."
- **Amount payable per installment:** The student indicated 500 shekels.
- **Payment frequency:** The student opted for monthly payments.

The student also provided supporting documents, including income statements, bank statements, and a detailed letter explaining their financial situation.

The financial department carefully reviewed the request and verified the submitted documents. After approval, the accountant developed a customized payment plan aligned with the student's financial condition. The plan outlined the following:

1. **Installment amounts:** Divided into manageable payments.
2. **Payment frequency:** Weekly or monthly, depending on the student's financial capacity.
3. **Payment start date:** To allow the student to organize their finances.

Once both parties approved the payment plan, the accountant sent a notification detailing the installment schedule and provided information about the available payment methods, such as cash or check.

Error:

- In the process of examining the students' tuition payments, the accountant observed that some students had not paid their tuition fees despite the end of the specified period, without any reason or any request submitted for the delay in payment. Therefore, the accountant sent a notice alerting the student that the specified payment period had ended, which would result in the school portal being temporarily suspended and he being given three days to pay his tuition. After these days, the student's registration will be deleted from the data archive, meaning that he will not be registered for the current semester.

Post-completion system state:

- The accountant has logged into the system. The data is accessed. The students who have paid the school fees and the students who have not paid are identified. Notifications are sent to the student according to his status. Adding payment information and changing the payment status to "paid" for paying students. Giving a specific deadline to non-paying students according to the status.

2- Scenario: Creating Teachers Schedules (Sadeel Jabareen):

Initial Assumption:

The principal accesses the scheduling portal using the teacher's User ID and full name. These preparations provide a good foundation for an efficient scheduling system that enables the principal to easily create the teacher's schedule.

Normal Flow:

1. The principal logs into the portal using their credentials (username and password) and navigates to the scheduling section. The interface provides a dashboard displaying an overview of the current scheduling status, enabling quick access to relevant information.
2. In the scheduling interface, the principal enters the teacher's **User ID** and **full name**. The system provides live recommendations by displaying matching names in a drop-down list, reducing errors in selecting the correct teacher.
3. After entering the teacher's details, the system validates the input and displays the teacher's information, including their current timetable, assigned classes, certifications, and any special notes (e.g., sick leave or preferred teaching times). This comprehensive view assists the principal in making informed scheduling decisions.

4. The principal reviews the teacher's schedule and begins assigning classes. A calendar view shows available time slots, allowing the principal to drag and drop classes into the teacher's schedule. This intuitive interface prevents overlapping assignments and simplifies the scheduling process.

5. The principal can choose between **AI-assisted scheduling** or **manual scheduling**. The AI considers factors such as the teacher's experience, previous class sizes, and student needs to recommend optimal assignments. Alternatively, the principal can manually adjust the schedule to align with the school's objectives.

6. As classes are assigned, the system automatically checks for conflicts, such as overlapping classes or double bookings. Any detected conflicts are flagged with visual indicators, prompting the principal to resolve them by adjusting timetables or reassigning teachers.

7. The system monitors the distribution of classes among teachers. If a teacher is assigned an excessive workload, the system notifies the principal to adjust the schedule, preventing potential burnout and ensuring equitable distribution of responsibilities.

8. Once all conflicts are resolved and workloads are balanced, the principal finalizes the schedule. The system then notifies teachers of their assignments via email, SMS, or in-portal messages, ensuring timely communication.

9. The system logs all scheduling activities, including the principal's User ID, timestamps, and actions taken. This audit trail ensures accountability and provides a reference for future scheduling processes.

Alternative Flow:

1. Resolving Scheduling Conflicts:

- If conflicts are detected (e.g., overlapping classes or double bookings), the system notifies the principal and provides detailed information about the issues. The principal can resolve conflicts by either adjusting the timetable or reassigning teachers to different classes.

2. Overriding AI-Generated Schedules:

- If the principal is dissatisfied with the schedules generated by the AI, they can manually override and modify the schedules. The system allows the principal to adjust input parameters, such as teacher availability or class sizes, and regenerate the schedule accordingly.

3. Addressing Workload Imbalances:

- The system continuously monitors teacher workloads. If it detects inequities (e.g., one teacher being overburdened), it alerts the principal. The principal can then redistribute classes to ensure a fair and balanced workload among teachers.

4. Final Review and Approval:

- Before finalizing the schedule, the principal conducts a final review to ensure all conflicts are resolved and workloads are balanced. Once satisfied, the principal confirms the schedule, locking it in for use.

5. Revised Notifications:

- If further modifications are made to the schedule after initial notifications are sent, the system automatically sends updated notifications to the affected teachers via email, SMS, or in-portal messages. This ensures all teachers are informed of their latest assignments.

Error Flow:**- Incorrect User ID Entry**

When the principal is inputting the teacher's User ID, if they input it wrong, then the system will give an error message saying to check the input.

- Database Errors

When there is a problem with a database, the system informs the principal and suggests that they may want to try again at a later time. This makes the principal be informed on any technical issue that may arise.

- AI Scheduling Failures

If the AI scheduling feature does not work as intended due to lack of complete data, the system notifies the principal to input the missing data before the system tries to create the schedule again.

- Notification System Issues

If there are issues with the notification system for example an email service provider then the system alerts this as a problem and recommends the use of other means of notifying the teachers.

Final State:

Upon completion of the scheduling process, the teacher's schedule is:

1. Finalized and Conflict-Free: All classes are block taught and students do not have any overlapping schedules.

2. Balanced in Terms of Workload: Teacher's workloads are well balanced so there is less of a chance of burnout.

3. Reviewed and Approved by the Principal: The principal has done a review of the schedule and all aspects of the schedule meet the needs of the school as per his observation.

3- Scenario: Publishing grades (Rand Amly):

Baseline Assumption: The semester has just started, a month into the semester. Here, it is now the time for the teachers to input and manage the grades in the portal.

Normal Flow

1. The teacher logs into the portal and navigates to the Grades section.

He/She enters the daily and monthly final exam grades, behavior and activity grades, and homework grades through the appropriate input fields.

The portal automatically saves these entries and displays confirmation.

When the teacher enters all daily exam grades including extra grades, the system automatically selects the top two grades in the test for each student and adds them to the total calculation.

Teachers can manually override the results if necessary to reflect specific considerations (e.g., technical issues during the test).

2. The teacher records the students' attendance using the "Attendance Tracker" icon.

In case of student absences, the system automatically sends notifications to parents about their children's absences via SMS or email.

If a student exceeds 15 days of unjustified absence in a semester, the system classifies the student as a failure and sends a notification to both the teacher and parents.

3. The portal system automatically calculates the semester average for the students when the teacher enters the grades and provides a means for the teacher to reweight the grades.

4. The student and parent will receive a notification each time the teacher posts a grade.

Alternative Flows

1. Grace Period for Grade Adjustments

- There is a provision for the teacher to modify grades for a certain grace period, say one week, in case of errors in data entry. Beyond that, all changes are routed for approval from the administration.

2. Behavior and Activities Marks

- Inbuilt AI within the portal would suggest behavior marks that may be tricky for a teacher to decide based on predefined parameters like class participation and punctuality.

3. Failure and Success Analysis

- Graphical representation of the performance analysis of the class showing success and failure rates. The system also suggests any changes required - curve for adjustment, if required.

Error flow:

1. System crash while feeding the input

- Most importantly, the data get lost if the system crashed while entering the marks unsaved data may be lost. But if the system gets rebooted, then it does produce an alert for the teacher and will have to rewrite the same lost data.

2. Incorrect Parents Notification

- In rare cases , the system might modify the wrong parents about a student's absence . if this occurs , the teacher can flag the error in the system , which automatically sends a correction to affected parents

Final Status:

1. Grades Posted Successfully: All grades related to exams, behavior, activities, and assignments have been posted.
2. Notifications Sent: To inform students and parents of grades.
3. Recording of Attendance and Absences: In case of an absence or failure reported, it is recorded and notified if necessary.
4. Class Performance Analysis: The performance is analyzed using AI, performance statistics, and suggestions when necessary are provided.
5. System Recording: The grade of each student is kept on record in the system together with attendance records for subsequent performance information.

4- Scenario: Creating a new account and assign role for a new user (Wasfia Awwad):

Normal flow:

The registration employee opens the "Create New Account" section in the portal system.

The system shows a form of account creation.

The registration employee enters in the user's essential information: User ID, Full Name, Personal Picture, Email, Phone Number, Password, Assigned Classes.

The registration employee adds additional personal information to the form which is the medical information (Allergies, Chronic diseases, Previous vaccinations, Medications, Previous medical operations).

The registration employee assigns the user a role (Student, Teacher, Accountant, etc.).

The employee submits the form.

The system validates the entered data and saves them in the user information records.

The system confirms that the account has been successfully created.

Alternative Flows:

1- User ID already assigned to another user

If the selected user ID already exists in the user information records, the system displays an error message: "A user with this ID already has an account. Please enter a different user ID."

The registration employee makes the edits the user ID and resubmits the form.

2- Missing mandatory fields

If any of the required information is missing, the system highlights the incomplete fields and displays a message: "Please fill all required fields."

Employee fills in the missed information and resubmits the form.

Error Flows:

1- System Times Out:

Registration employee fills out the user account form, then submits it.

The portal system will try to process and store info, because the portal system times out due to server overloaded/ Network problems.

The portal system pops an error message "Unable to process your request right now. Please try again later."

The register employee won't be able to continue further and has to wait until the problem is solved or call technical support.

2- User loses all password recovery options:

The user forgets their password and attempts to reset their password but realizes they no longer have access to their registered email, or phone number.

The user is unable to proceed further and must visit or contact the registration department.

The registration employee manually verifies the user's identity using official identification documents or other records.

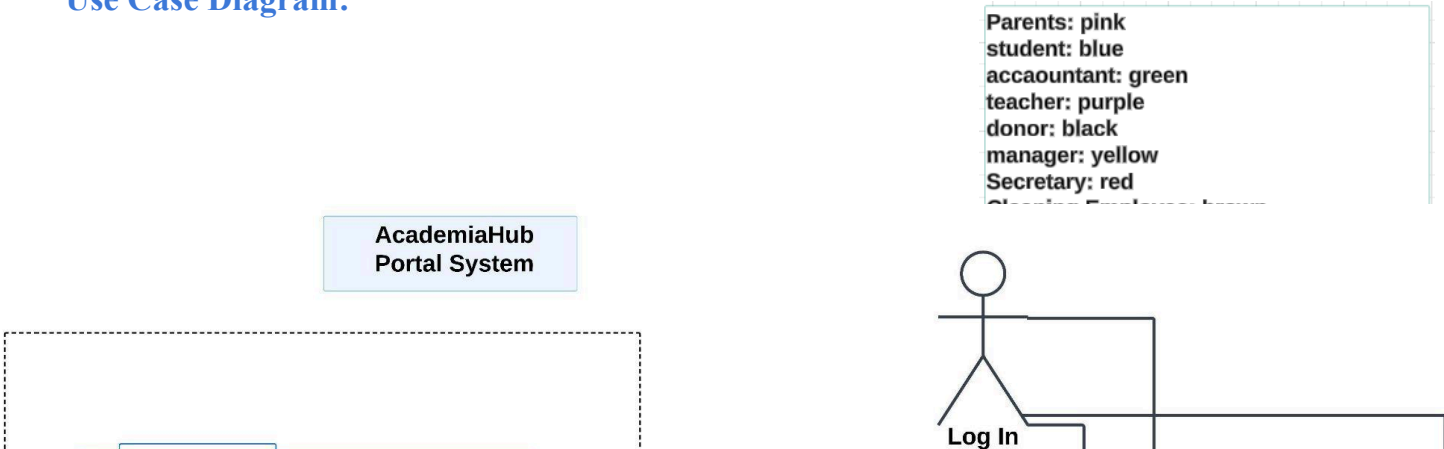
In case of verification, the registration employee resets the password and updates the recovery options.

If verification fails, then the account remains locked and the issue is escalated to technical support for further review.

Final Status:

The user account is fully created, and all provided information is stored in the portal system records. The registration employee receives a confirmation message, and the user can now log in to their account using their assigned ID and password.

Use Case Diagram:



Use cases description:

- **Actors:**

- Parents.
- Student.
- Teacher.
- Donor.
- Accountant.
- Secretary.
- Registration Employee.
- Principal.
- Education Ministry.
- Cleaning Employee.
- SMS System.
- Email System.

- **Descriptions:**

Parents:

1. View financial record: Parents can see all financial transactions regarding their child, like tuition fees and donations.

2. Get warnings and alerts: parents receive notifications on academic warnings, attendance issues, or any other important update.

3. Make son evaluations: The parent makes an assessment of his child about their academic performance to let him know his shortcomings.

4. View payment: Parents are allowed to see payment schedules and remaining balances.

5. Be in chat groups: Parents can be added into groups where other parents also have children in the same class, to communicate and share information.

- 6. View teacher office hours:** View and schedule meetings with teachers about your child during their office hours.
- 7. Send payment reminder:** Receive notifications on when the payments are due, or when they are overdue.
- 8. View schedules:** The parents can access their child's academic schedule to keep track of classes.
- 9. Send messaging:** Parents can send messages to teachers, staff, or other parents directly through the portal.
- 10. View messaging:** Parents can receive and read messages from teachers, staff, or other parents.

Teacher:

- 1. Split student:** Teachers are able to assign or move students into different groups or classes when needed.
- 2. Publish learning materials:** The teacher can upload resources such as assignments, study materials, and notes for the students.
- 3. View salary:** Provides teachers with their salary details, even the status of salary deposits.
- 4. Get salary notification:** Sends notification to the teacher once a salary is deposited.
- 5. Publish assignment:** Assigns assignments to students and also gives the deadline for submissions.
- 6. Publish grades:** Teachers can upload and share grades of students in every class or subject.
- 7. View teacher office hours:** A teacher can view and manage their office hours to student meetings.

- 8. View schedule:** View own teaching schedule and class timings.
- 9. Send messaging:** Teachers can send messages to the students, parents, or school staff.
- 10. View messaging:** Teachers can read incoming messages from the students, parents, or staff.

Student:

- 1. Make Discount:** Students may apply for a discount depending on their academic performance or need.
- 2. Assignment submission:** Students can submit their assignments by uploading to their respective teachers.
- 3. Learning Material View:** The students can view and revise study materials and other resources provided by the teachers.
- 4. Payment:** Allows the student to pay fees regarding school, etc.
- 5. Grades View:** In this feature, students are allowed to view their grades so that they can keep up with their performance at school.
- 6. Evaluation of Teacher:** Students can evaluate the performance and teaching capability of a teacher.
- 7. Add subtitles to audio:** Adding subtitles to any education-related audio to facilitate easy understanding by students.
- 8. Audio of texts:** The facility to hear texts is provided for accessibility.
- 9. Test games for practice and learning:** Several test games can be staged to test the knowledge for better learning.
- 10. Advise on field selection:** Students of higher classes will be able to get advice about their probable academic track/field.

11. Request Electronic Documents: Transcripts or Certificate requests can be raised electronically.

12. View schedules: The students can view class schedules and events connected with their classes.

Donor:

1. Make financial donation: Donors can provide financial donations to the school or some of its programs.

2. Make non-Monetary donation: Donors can donate in-kind, such as supplies, books, or other materials.

Accountant:

1. Receive financial donation information: Accountants receive information regarding the financial donations that were provided by either parents or donors.

2. Make a payment: Accountants can make a payment; examples include tuition and other fees.

3. Salaries deposit: Accountants are responsible for depositing salaries for staff and teachers within the school.

4. Send donation info: Accountants send out any information concerning donations to higher relevant authorities such as the Ministry of Education.

5. Create financial report: Accountants create reports detailing the financial status of the school.

6. View grades: The accountant may be given access to view grades, which helps in providing student financial aid and scholarships.

7. Check payment: Accountants can check the status of the payments made by students or parents.

8. View financial status and personal info: Accountants can view the financial status and other related personal information of students or staff.

Secretary:

1. Receive non-Monetary donations: Secretaries process the non-monetary donations such as books and uniforms.

2. Managing books and uniforms: Secretaries manage the distribution of books and uniforms to students.

3. Arrange meetings for the principal: Secretaries arrange meetings for the principal with the staff, parents, or any other stakeholder.

4. Send donation information: Secretaries send information regarding donations to the concerned individuals or departments.

Registration Employee:

1. Account creation and assigning roles: The registration employee creates accounts for students, teachers, and staff and assigns them their respective roles.

2. Student transfer: The registration employee is responsible for the transfer of students in and out of the school.

3. Transfer teacher: Registration clerks are responsible for transferring teachers to other departments or schools.

4. Generate student's electronic documents: The registration clerks generate students' electronic documents, such as transcripts or certificates.

Principal:

- 1. Salary deduction:** The manager in charge can approve the deduction of salary for the staff based on performance or attendance.
- 2. View donation information:** Managers can access information about the donations made to the school.
- 3. Donation info send:** Managers share information about donations with concerned authorities.
- 4. Student report monitor:** Managers can monitor reports of students' performance for the smooth running of affairs.
- 5. Teacher report monitor:** Managers can monitor reports of teachers' performance and take necessary actions.
- 6. Create schedule:** Managers create and maintain schedules regarding teachers and students.

Education Ministry:

- 1. Donation info view:** The Education Ministry will be able to view information about donations for transparency and to manage funds.

Cleaning Employee:

- 1. Task list creation:** The cleaning employees are given the list of places to be cleaned.
- 2. Area Allotment:** Areas to be cleaned are assigned to cleaning employees. A person after cleaning an area can mark the area.

Login User:

1. **Forget Password:** A user clicks to indicate they have forgotten their password and initiates some sort of password recovery process.
2. **Reset Password:** User resets the password successfully after identification through the SMS or email notification.

SMS System:

1. **Send SMS Notification:** An SMS is sent to the user with a link or verification code that will assist in resetting the password.

Email System:

1. **Send Email Notification:** An e-mail will be sent to the user containing a link or code for secured password resetting.

Task 3.3: [Individual Task]:

AcademiaHub System

Use-Case: *Student payment management*

Version 2.0

Revision History

Date	Version	Description	Author
25/Sep/2024	Draft	Draft version	Toqa Abdeen
19/Oct/2024	Version 1.0	Adjustments made following the review.	Toqa Abdeen
23/Nov/2024	Version 2.0	Amend the section pertaining to use case extensions. Conduct a final review and cleanup. Integrate alternative flow scenarios. Resolve outstanding issues.	Toqa Abdeen

1.1 Brief Description

This exceptional tool facilitates the management of school finances for the entire student body.

1.2 Actor

1.2.1 The actors for this use case are the Accountant and Student.

2. Preconditions (Entry condition)

2.1 Log In

Before this use case begins the Actors have logged onto the system.

2.2 Record Access

The Accountant actor has to access the system's data archive.

3. Flow of Events

The use case commences when the Accountant actor clicks on the "Manage Tuition Fees" feature from the Main Form.

3.1 Basic Flow - <Managing Tuition Fees>

- 3.1.1 The accountant selects "Manage Tuition Fees" from the system menu.
- 3.1.2 The system prompts the accountant to "Enter Student ID".
- 3.1.3 The accountant enters the student's assigned number under "Enter Student ID".
- 3.1.4 The system verifies the student's number and displays the corresponding tuition fees under "Verify Student Details".
- 3.1.5 The accountant selects the appropriate tuition fee from "Choose Tuition Fee".
- 3.1.6 The system guides the accountant through the following steps:
 - a. Input the Amount of Installment Paid.
 - b. Input the Date of Payment.
 - c. Select the Method of Payment (Cash/Check).
 - d. If the method of payment is by check, enter the Number of Checks.
 - e. Generate and review a Payment Acknowledgment Receipt
- 3.1.7 The accountant selects "Save" to finalize the student's financial information.
- 3.1.8 The system updates the student's tuition payment status to "Paid" and confirms the changes.

4. Alternative Flows

- 4.1** School fees remain outstanding and request for an extension of the payment deadline has been submitted.

The application will be reviewed by the accountant and the financial department. Based on their evaluation, a notification will be sent indicating whether it has been approved or rejected.

5. Post-conditions (Exit condition)

- 5.1** The system's financial data repository is updated to reflect the latest tuition fee entries.
- 5.2** The system returns to the main menu or the previous screen, ready for the next operation.

6. Special Requirements

- 6.1** The system shall process and update the tuition fee records in less than 5 seconds after the "Save" option is selected.
- 6.2** The system shall retrieve the student's details and corresponding tuition fees in less than 3 seconds after entering the student ID.
- 6.3** The system shall ensure that the selected tuition fee is accurate and corresponds to the correct student based on the student ID.
- 6.4** The system shall provide a report of all transactions completed in a given period, accessible by authorized personnel only.

7. Extension Points

- 7.1** Make discounts use case.
- 7.2** Cancel student registration use case.

AcademiaHub System

Use-Case Specification

Use-Case: *Teacher Publishing Grades*

Version 2.0

Revision History

Date	Version	Description	Author
25/Nov/2024	Draft	Initial draft version	Rand Amly
30/Dec/2024	Version 1.0	Adjustments made following the review.	Rand Amly
10/Jan/2025	Version 2.0	Added alternative and error flows, finalized draft	Rand Amly

1.1 Brief Description

This use case describes how teachers will input, manage, and publish student grades , keep track of attendance, and analyze performance with AI-powered suggestions.

1.2 Actor(s)

1.2.1 Primary Actor: Teacher

1.2.2 Secondary Actor: Student, Parent

2. Preconditions (Entry Conditions)

2.1 The teacher should be logged into the system.

2.2 The teacher needs to have access to Grade Entry and Attendance Management of the system.

3. Flow of Events

3.1 Basic Flow: Grade Entry and Publication

3.1.1 The teacher logs into the system and selects the "Grade Entry" section.

3.1.2 The system displays an input form for grades, whether daily, monthly, and final exams.

3.1.3 In it, the teacher should enter grades concerning behavior, activities, and assignments.

3.1.4 The semester average is computed automatically by the system.

3.1.5 The system will calculate the highest two exams out of all automatically

3.1.6 A teacher may edit the weight of certain grades manually if so wished.

3.1.7 Store grades and notify students/ parents about them.

3.2 Attendance and Absence Management

3.2.1 Teacher will mark the attendance of the student by using "Attendance tracker".

3.2.2 If a student is absent then the system automatically sends notifications to parents.

3.2.3 Students who have exceeded more than 15 days without valid reasons get marked for failure by the system, and in turn it notifies the teacher and parents of that particular student.

3.3 AI Powered Performance Analysis

3.3.1 System analyzes the class performance and prepares success and failure rates.

3.3.2 If needed, the system should provide recommendations such as curve adjustment to have a better grade distribution.

4. Alternate Flows

4.1 Grade Input Error

4.1.1 Grades shall be edited by the teacher if there was some error on his/her part
4.1.2 while putting in grades within the available grace period of time.

4.2 Behavioral Marks Assistance

4.2.1 The AI suggestion facility shall be used by the teacher to support the
4.2.2 suggestion of behavior marks based on participation data.

5. Postconditions (Exit Conditions)

5.1 Grades and attendance records are kept in the system.

5.2 Notifications to be sent to the students and parents.

5.3 The teacher shall be taken back to the menu for further actions.

6. Special Requirements

6.1 The system should store grades and attendance data within less than 3 seconds from the time of submission.

6.2 Notification to parents and students shall be sent within 2 seconds of publishing the grade.

6.3 The grading reports shall be accessible only to the authorized users through the system.

7. Extension Points

7.1 "Add Extra Credit Marks" Use Case

7.2 "Override Absence Notifications" Use Case

AcademiaHub System

Use-Case Specification

Use-Case: *Creating Teacher Scheduling*

Version 2.0

Revision History

Date	Version	Description	Author
15/ Dec/2024	Draft	Initial draft created	Sadeel Jabareen
19/ Dec/2024	Version 1.0	Enhanced the document with additional details, introduced new processes, and expanded the special requirements section.	Sadeel Jabareen
25/ Dec/2024	Version 2.0	Modified the section on use case extensions, completed final refinements, added alternative processes, and resolved remaining issues.	Sadeel Jabareen

1.1 Brief Description:

This use case makes it easy for a school principal to be able to send teachers to classes so that there is no conflict between classes, as well as ensure that the workload is divided fairly. The assignments go through rule checks and teachers are immediately notified of their new schedules. Also, the system incorporates the features of an audit trail and has AI-enabled features for scheduling.

1.2 Actor:

1.2.1 Principal According to the scheduling portal the teachers are allocated classrooms by the Principal. They will either have to assign teachers individually or use AI suggestions to create the most optimal schedules.

2. Preconditions (Entry condition):

2.1 To view and update results, The Principal has to log in the system with proper identification information.

2.2 The list of teachers and their schedules, as well as the schedules of classes, is to be up to date and available.

2.3 The scheduling module has to be working and serve as a component of the school management system.

3. Flow of Events:

3.1 Basic Flow - Assign Teacher:

3.1.1 The principal logs into the scheduling application and selects the “Assign Teacher Schedule ” option.

3.1.2 The system displays a list of available teachers, including their current schedules, workloads, and any posted regulations or preferences (e.g., certifications, preferred teaching times).

3.1.3 The principal inputs the teacher’s **User ID** or **Name** to select the teacher.

3.1.4 The principal selects the class name, timetable, and subject for the teacher.

3.1.5 The system performs the following validations:

- **Conflict Check:** Ensures the teacher is not assigned to another class at the same time.
- **Quota Check:** Verifies the teacher’s assigned classes do not exceed the allowed quota.
- **Workload Balance:** Ensures the teacher’s workload is reasonable compared to others.

3.1.6 **System Response** (If all validations pass):

- The system updates the teacher's TIMETABLE.
- The system sends a real-time notification (email, SMS, or in-app message) to the teacher about the new assignment.
- The system logs the action (including timestamp and principal's User ID) for the audit trail.

3.1.7 The process concludes once the assignment is successfully completed.

4. Alternative Flows:

4.1 Teacher Already Assigned

1. Trigger: During **Step 3.1.5 (Conflict Check)** in the Basic Flow, if the selected teacher is already assigned to another class at the chosen time.

2. System Response:

- 4.1.1 The system displays a conflict message.

3. User Action:

- 4.1.2 The principal can:
 - Adjust the timetable (e.g., reschedule the class).
 - Assign a different available teacher.

4. Return to Basic Flow: After resolving the conflict, the system returns to **Step 3.1.5** to revalidate the assignment.

4.2 Quota Exceeded

1. Trigger: During **Step 3.1.5 (Quota Check)** in the Basic Flow, if the teacher's assigned classes exceed the allowed quota.

2. System Response:

- 4.2.1 The system generates a warning message.

3. User Action:

- 4.2.2 The principal can:
 - Reassign the class to another teacher.
 - Manually override the quota (if permitted).

4. Return to Basic Flow: After resolving the quota issue, the system returns to **Step 3.1.5** to revalidate the assignment.

4.3 AI-Assisted Scheduling

1. Trigger: At any point during **Step 3.1.4 (Class Assignment)** in the Basic Flow, the principal chooses to use AI-assisted scheduling.

2. User Action:

- 4.3.1 The principal enables the AI scheduling feature.

3. System Response:

- 4.3.2 The system processes data (e.g., teacher experience, class size, student needs) and generates an optimized schedule.

- 4.3.3 The system displays the recommended schedule for the principal's approval.

4. Return to Basic Flow: After the principal approves the AI-generated schedule, the system proceeds to **Step 3.1.5** for validation.

4.4 Notification Delivery Failure

1. Trigger: During **Step 3.1.6 (Notification)** in the Basic Flow, if the system fails to deliver a notification.

2. System Response:

- 4.4.1 The system retries sending the notification up to three times.
- 4.4.2 If delivery still fails, the system logs the error and notifies the principal.

3. Return to Basic Flow: After resolving the notification issue, the system completes **Step 3.1.6** and ends the use case.

4.5 Changing the Load Distribution

1. Trigger: During **Step 3.1.5 (Workload Balance)** in the Basic Flow, if the system detects an imbalance in workload distribution.
2. System Response:
 - 4.5.1 The system sends a notification to the principal with suggestions for rebalancing.
3. User Action:
 - 4.5.2 The principal can:
 - Reassign classes to balance workloads.
 - Approve the imbalance with a justification.
4. Return to Basic Flow: After resolving the imbalance, the system returns to **Step 3.1.5** to revalidate the assignment.

4.6 Changing Current Schedules

1. Trigger: After Step 3.1.6 (Successful Assignment) in the Basic Flow, if a schedule needs to be modified (e.g., due to teacher absence).
2. User Action:
 - 4.6.1 The principal updates the timetable by reassigning classes or teachers.
3. System Response:
 - 4.6.2 The system sends updated notifications to affected teachers and logs the changes.
4. Return to Basic Flow: After updating the schedule, the system returns to Step 3.1.5 to revalidate the assignment.

5. Post-conditions (Exit condition):

- 5.1 The teacher's schedule is generated and saved with no overlapping of classes.
- 5.2 Teachers receive real-time notification on new assignments or changes made to the current ones.

5.3 All scheduling actions are recorded in order to provide an audit trail.

6. Special Requirements:

6.1 Performance The system must be able to handle scheduling requests and present the results in not more than 2 seconds.

6.2 Notification effectiveness Notifications must be fulfilled with a reliability of 99,9 percent and should reach the target recipients in 5 seconds.

6.3 Security The system has to incorporate means for identification for authentication for securing of sensitive data.

6.4 Auditing The system also must be capable of tracking all possible actions executed in regards to the schedule with necessary comments, timestamp, and user data.

6.5 Ease of Integration The integration of the scheduling module to other systems in the school like attendance, teacher's availability calendar etc. must be easy.

7. Extension Points:

7.1 Integration with Teacher Availability Current service provides for a real time availability which can notify the prospective students with teachers' availability.

7.2 Advised computer-based insights for future development are in conflict identification also as workload estimation.

AcademiaHub System

Use-Case Specification

Use-Case:*Creating a new account and assign role for a new user*

Version 2.0

Revision History

Date	Version	Description	Author
20 Dec/2024	Draft	Initial draft created	Wasfia Awwad
21/ Dec/2024	Version 1.0	Modified the use case section, added alternative processes, error processes and final state.	Wasfia Awwad
25/ Dec/2024	Version 2.0	added to the document additional details, updated diagrams.	Wasfia Awwad

1.1 Brief Description

This use case enables registration employees to create accounts for users by filling out essential and additional personal information, assigning roles, and ensuring all details are stored accurately. It also supports changing the password in case a user forgets it by providing a secure process for password retrieval.

1.2 Actor

1.2.1 Registration Employee

Preconditions (Entry condition)

2.1 The registration employee must log into the portal system.

2.2 The system must be operational, and all user records must be accessible for validation.

Flow of Events

3.1 Basic Flow - Create User Account

3.1.1 The registration employee selects the “Create Account” option in the portal to go to the “Create Account” section.

3.1.2 The employee fills out the user’s essential information, including user ID, full name, email, password, and assigned classes.

3.1.3 The employee fills additional personal information, which are the medical details (allergies, chronic diseases, etc.).

3.1.4 The employee assigns a role to the user (e.g., student, teacher, parent).

3.1.5 The portal system validates the entered information, ensuring no duplication (e.g., unique user ID).

3.1.6 If validation succeeds:

The system creates the user account.

A confirmation message is displayed.

The use case ends.

Alternative Flows

4.1 User ID already used.

4.1.1 If the user ID is already in use, the system displays an error message.

4.1.2 The registration employee edits the form and resubmits it.

4.2 Missing Mandatory Fields

4.2.1 If any mandatory field is missing, the system highlights the missing fields and displays a warning message.

4.2.2 The registration employee completes the missing information and resubmits the form.

Post-conditions (Exit condition)

5.1 The user account is successfully created and saved in the system accounts records.

5.2 The new user can log in using their ID and password.

5.3 All registration actions are logged for auditing purposes.

Special Requirements

6.1 The system must validate user information within 2 seconds.

6.2 Notifications for successful account creation must be sent to the user's email address immediately.

6.3 The system must ensure secure handling of sensitive data, such as medical information.

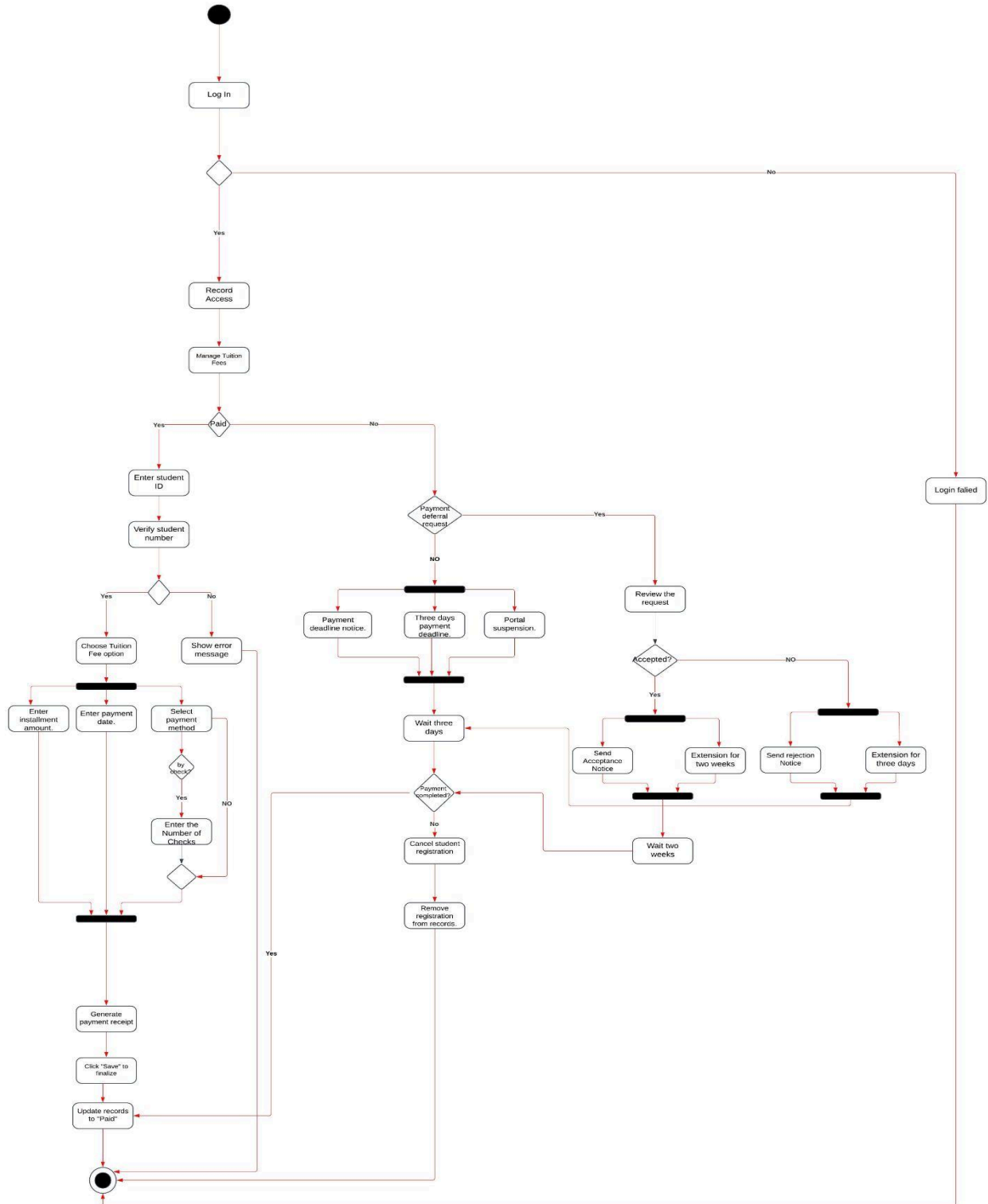
Extension Points

7.1 Integration with a notification system for sending account creation confirmation.

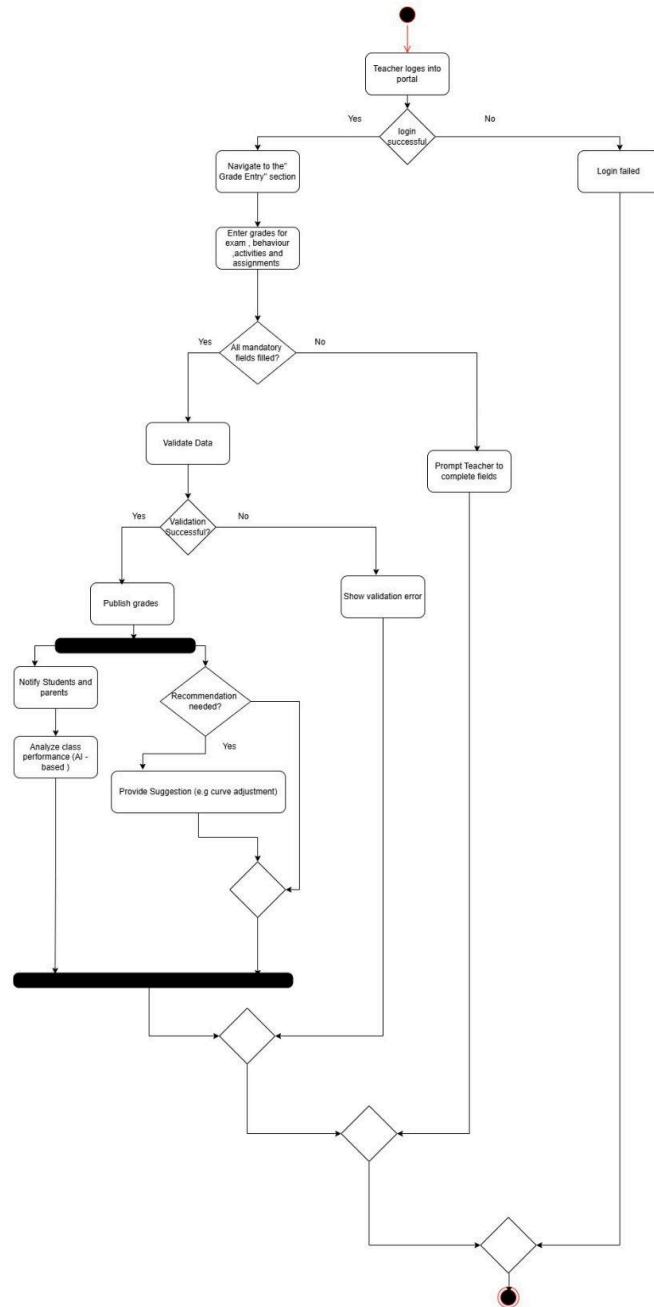
7.2 Integration with SMS and email services to deliver new passwords to users in case they forget their current password.

Task 3.4: [Individual Task]:

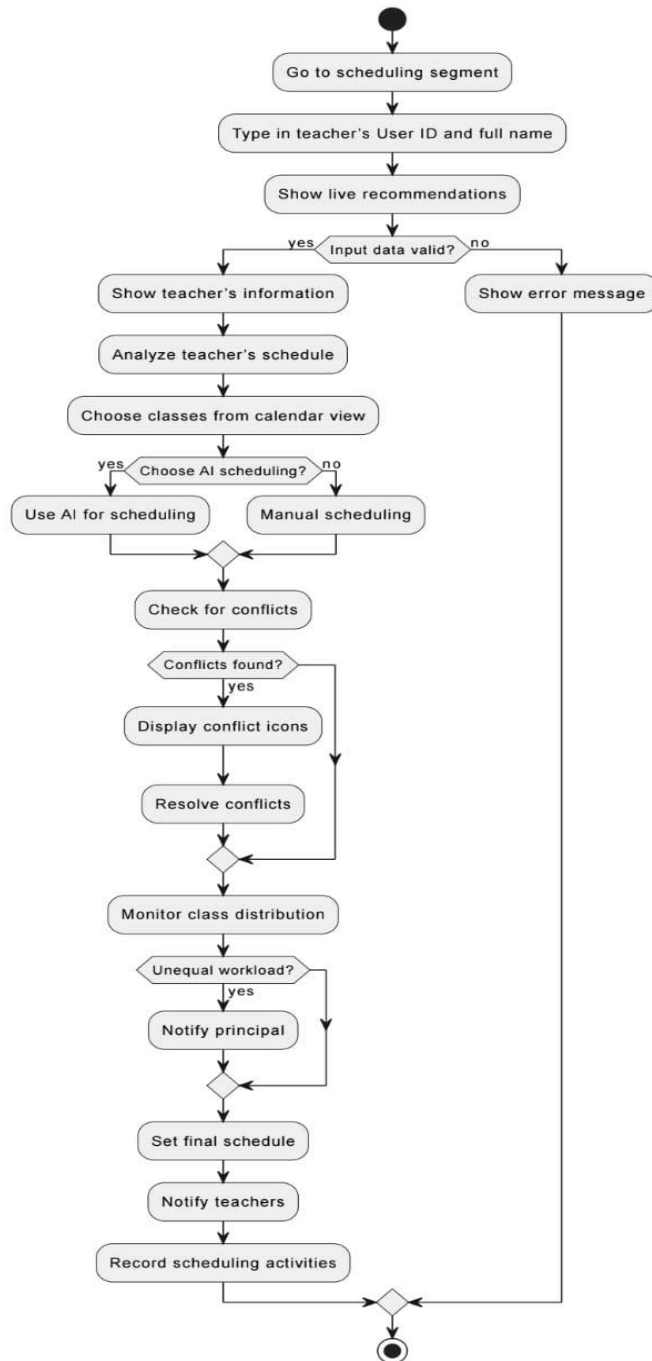
Student payment management Activity Diagram (Toqa Abdeen) :



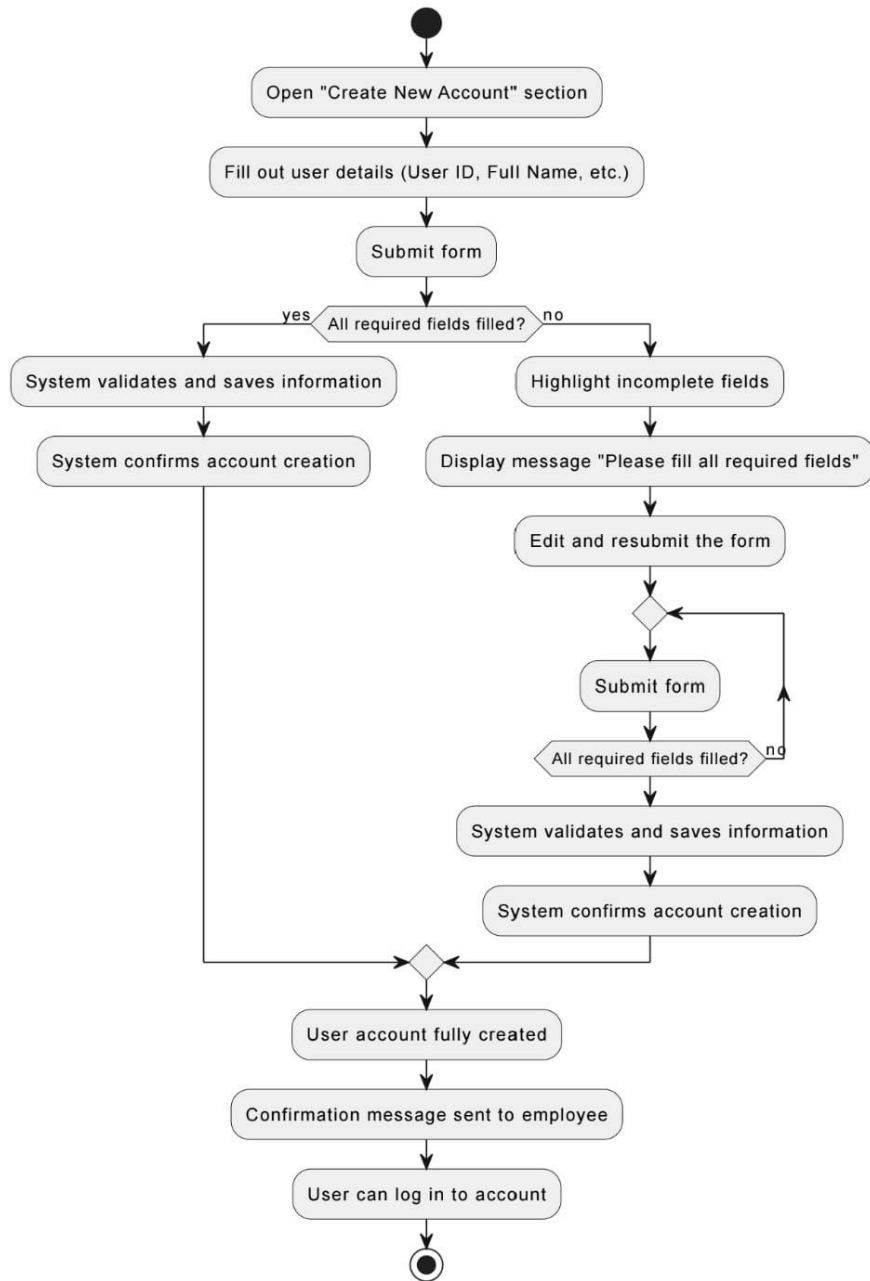
Publishing grades Activity Diagram (Rand Amly):



Creating Teacher Scheduling Activity Diagram (Sadeel Jabareen):



Creating a new account and assign role for a new user (Wasfia Awwad):



Task 4.1: [Group Task] System Class modelling and Analysis:



1. Teacher

This class represents the teachers in the system.

- **Attributes:**
 - TeacherID (int): Unique identifier for the teacher.
 - Name (String): Name of the teacher.
 - Email (private String): Email address of the teacher.
 - PhoneNumber (private String): Contact number of the teacher.
 - ClassSection (String): The class section assigned to the teacher.
 - salary (private double): Salary of the teacher.
 - officeHours (LocalTime): Office hours availability of the teacher.
 - medicalInfo (MedicalInfo): Medical details of the teacher.
 - **Operations:**
 - Publishing learning materials and assignments.
 - Managing schedules and viewing salary.
 - Sending messages and publishing grades.
 - Viewing and managing office hours.
-

2. Secretary

This class represents the secretary of the school.

- **Attributes:**
 - SecretaryID (private int): Unique identifier for the secretary.
 - SName (String): Name of the secretary.
 - SEmail (String): Email address of the secretary.
 - SPhoneNum (long): Contact number of the secretary.
 - NumOfStudents (private int): Number of students managed by the secretary.
 - **Operations:**
 - Arranging meetings for the principal.
 - Managing books and uniforms.
 - Sending donation-related information.
-

3. RegistrationEmployee

This class manages the registration of students and teachers, including their role allocation.

- **Attributes:**
 - registerID (private int): Unique identifier for the registration employee.
 - password (private String): Login password for the employee.
 - reportingTo (String): The supervisor for the employee.
- **Operations:**

- Creating accounts for students and teachers.
 - Transferring students or teachers between roles or locations.
 - Assigning roles and generating documents.
-

4. CleaningEmployee

This class manages cleaning activities in school areas.

- **Attributes:**
 - EmployeeID (private int): Unique identifier for the cleaning employee.
 - password (private String): Login password for the employee.
 - ShiftSchedule (String): Work schedule of the cleaning employee.
 - **Operations:**
 - Creating a list of cleaning tasks.
 - Adding or removing cleaning tasks.
 - Clearing all tasks.
-

5. Parent

This class represents the parents of students in the system.

- **Attributes:**
 - ParentID (private int): Unique identifier for the parent.
 - PName (String): Name of the parent.
 - PEmail (String): Email address of the parent.
 - ParPhoneNum (long): Contact number of the parent.
 - Other private fields include ParentsPlaceOfResidence, ParentsPlaceOfWork, and ParentsMaritalStatus.
 - **Operations:**
 - Viewing financial records and payment details.
 - Receiving alerts and reminders.
 - Sending evaluations and messages to teachers.
 - Joining chat groups for their child's class section.
-

6. LoginUser

This class handles user authentication and login operations.

- **Attributes:**
 - UserID (String): Login ID for the user.
 - UserPassword (String): Password for user login.

- UEmail (String): Email address of the user.
 - UPhoneNum (long): Contact number of the user.
 - loginDate (LocalTime): Date and time of the user's last login.
 - **Operations:**
 - Login and logout functionality.
 - Resetting passwords.
-

7. Principal

This class manages overall school operations.

- **Attributes:**
 - PrincipalID (int): Unique identifier for the principal.
 - Name (String): Name of the principal.
 - Email (String): Email address of the principal.
 - PhoneNumber (String): Contact number of the principal.
 - salary (private double): Salary of the principal.
 - teacher (ScheduleTeacher): Schedule details for teachers.
 - **Operations:**
 - Managing salaries, donations, and reports.
 - Monitoring student and teacher reports.
 - Creating teacher schedules.
-

8. Accountant

This class is responsible for the financial management of the school.

- **Attributes:**
 - accountantID (int): Unique identifier for the accountant.
 - accountantName (String): Name of the accountant.
 - accountantEmail (String): Email address of the accountant.
 - accountantPhoneNumber (long): Contact number of the accountant.
 - **Operations:**
 - Managing donations and salaries.
 - Generating financial reports.
 - Applying discounts based on marital status.
-

9. ScheduleTeacher

This class represents the schedule of teachers.

- **Attributes:**
 - scheduleID (int): Unique identifier for the schedule.
 - classSection (String): Assigned class section.
 - startTime (LocalTime): Start time of the class.
 - endTime (LocalTime): End time of the class.
 - dayOfWeek (String): Day of the week for the schedule.
 - **Operations:**
 - Creating, updating, and viewing teacher schedules.
-

10. Student

This class represents students in the system.

- **Attributes:**
 - studentID (int): Unique identifier for the student.
 - studentName (String): Name of the student.
 - studentEmail (private String): Email address of the student.
 - phoneNumber (private String): Contact number of the student.
 - academicYear (int): Current year of study.
 - courses (Course[]): List of courses taken by the student.
 - **Operations:**
 - Submitting assignments and paying fees.
 - Viewing grades and accessing materials.
-

11. Course

This class represents academic courses in the school.

- **Attributes:**
 - courseID (int): Unique identifier for the course.
 - courseName (String): Name of the course.
 - assignments (Assignment[]): List of assignments for the course.
 - evaluation (char): Evaluation type for the course.
 - **Operations:**
 - Calculating grades based on assignments.
-

12. Assignment

This class represents tasks assigned to students.

- **Attributes:**
 - assignmentID (int): Unique identifier for the assignment.

- assignmentName (String): Name of the assignment.
 - startDate (Date): The start date of the assignment.
 - endDate (Date): The end date of the assignment.
 - grade (double): The grade received for the assignment.
 - evaluation (char): Evaluation type of the assignment.
 - **Operations:**
 - Getters and setters for assignment details.
-

13. StudentMarkSheet

This class represents the marks and performance assessments of a student.

- **Attributes:**
 - courses (Course[]): List of courses taken by the student.
 - assessmentBehaviour (char): Behavior evaluation of the student.
 - **Operations:**
 - getCourses() / setCourses(courses: Course[]): Retrieves or sets the list of courses.
 - getAssessmentBehaviour() / setAssessmentBehaviour(assessmentBehaviour: char): Gets or sets the behavior evaluation.
 - calculateAverage(): Calculates the average marks across all courses.
 - createDocument(): Generates a marksheet document.
-

14. Payment

This class is responsible for dealing with payment information.

- **Attributes:**
 - PaymentNum (int): Unique number for the payment.
 - PaymentAmount (double): Amount paid in dollars.
 - PaymentDate (Date): Date of the payment.
 - EndOfPayment (Date): Final payment due date.
 - PaymentMethod (String): Payment method used (e.g., credit card, cash).
 - Paid (double): Amount already paid.
 - **Operations:**
 - calcPayment(surplus, paid, amountDue): double: Calculates the remaining payment.
-

15. Classroom

This class represents a classroom.

- **Attributes:**
 - ClassSection (String): The section of the class.
 - StudentsNum (int): Number of students in the classroom.
 - **Operations:**
 - getClassSection() / setClassSection(section: String): Retrieves or updates the class section.
 - getStudentNum() / setStudentNum(num: int): Retrieves or updates the number of students.
-

16. MedicalInfo

This class represents medical information for individuals.

- **Attributes:**
 - ID (int): Unique identifier for the medical record.
 - allergies (String[]): List of allergies.
 - chronicDiseases (String[]): List of chronic diseases.
 - previousVaccinations (String[]): Record of past vaccinations.
 - medications (String[]): List of medications.
 - previousOperations (String[]): History of surgeries.
-

17. Message

This class manages messages between users in the system.

- **Attributes:**
 - messageTitle (String): Title of the message.
 - messageType (String): Type of the message (e.g., notification, personal).
 - messageContent (String): Content of the message.
 - senderID (int): ID of the message sender.
 - receiverID (int): ID of the message receiver.
 - sendingTime (LocalTime): Time the message was sent.
 - archivingTime (LocalTime): Time the message was archived.
 - messageID (int): Unique identifier for the message.
 - **Operations:**
 - Getters and setters for message attributes.
 - Validating, deleting, and updating messages.
-

18. ChatGroups

This class manages group chats for parents and teachers.

- **Attributes:**
 - parent (Parent): Parent associated with the group.
 - message (String): Content of the group messages.
 - **Operations:**
 - SendMessages(ParentName: String): Sends a message.
 - ReceiveMessages(message: String, ParentName: String): Receives messages in the group.
-

19. SMS_System

This class allows sending notifications via SMS.

- **Attributes:**
 - SMSID (int): Unique identifier for the SMS.
 - PhoneNumber (String): Recipient's phone number.
 - **Operations:**
 - sendSMSNotification(notificationMessage: String, phoneNumber: String): Sends an SMS notification.
-

20. Area

This class represents a physical area within the school.

- **Attributes:**
 - areaID (int): Unique identifier for the area.
 - areaName (String): Name of the area.
 - isCleaned (boolean): Cleaning status of the area.
 - lastCleanedBy (String): Name of the last cleaning employee.
 - lastCleanedTime (LocalDateTime): Timestamp of the last cleaning.
 - **Operations:**
 - highlightArea(employeeName: String): Highlights areas needing cleaning.
 - resetCleaningStatus(): Resets the cleaning status.
-

21. Donor

This class represents donors contributing to the school.

- **Attributes:**
 - donorID (int): Unique identifier for the donor.
 - donorName (String): Name of the donor.
 - donorEmail (String): Email address of the donor.

- donorPhoneNumber (long): Phone number of the donor.
 - **Operations:**
 - Getters and setters for donor attributes.
 - makeDonation(Donation: Donation): Processes a donation.
-

22. FinancialDonation

This class handles financial donations.

- **Attributes:**
 - amount (double): Amount donated.
 - **Operations:**
 - Getters and setters for the donation amount.
 - display(): Displays donation details.
-

23. NonMoneyDonation

This class manages non-financial donations.

- **Attributes:**
 - itemDescription (String): Description of donated items.
 - **Operations:**
 - Getters and setters for item descriptions.
-

24. Document (Interface)

This interface represents a document in the system.

- **Attributes:**
 - documentID (int): Unique identifier for the document.
 - documentTitle (String): Title of the document.
 - documentType (String): Type of the document.
 - releaseDate (Date): Date the document was released.
 - **Operations:**
 - readDocument(studentID: int): String: Reads the document content.
 - updateDocument(studentID: int, content: String): Updates the document.
 - deleteDocument(studentID: int, academicYear: int): Deletes the document.
-

25. EducationMinistry

This class represents the education ministry overseeing the school.

- **Attributes:**
 - ministryID (String): Unique identifier for the ministry.
 - ministryName (String): Name of the ministry.
 - ministryEmail (String): Email address of the ministry.
 - **Operations:**
 - viewDonationInfo(Principal: principalName): View donation-related information.
-

26. ScheduleStudent

This class manages student schedules.

- **Attributes:**
 - scheduleID (String): Unique identifier for the schedule.
 - classSection (String): Class section of the student.
 - startTime (LocalTime): Start time of the schedule.
 - endTime (LocalTime): End time of the schedule.
 - dayOfWeek (String): Day of the week for the schedule.
 - **Operations:**
 - createSchedule(studentID: Int, classSection: String, course: Course, startTime: LocalTime, endTime: LocalTime, dayOfWeek: String): void: Creates a schedule for a student.
 - updateSchedule(scheduleID: String, newStartTime: LocalTime, newEndTime: LocalTime): void: Updates an existing schedule.
 - deleteSchedule(scheduleID: String): void: Deletes a student schedule.
 - viewSchedule(studentID: String): List<Schedule>: Views the schedule for a specific student.
-

27. FinancialReport

This class represents a detailed financial report.

- **Attributes:**
 - payment (Payment): Payment details included in the report.
 - **Operations:**
 - getPayment() / setPayment(payment: Payment): Retrieves or updates payment information.
 - createDocument(content: String): void: Creates the financial report in a document format.
-

28. EmployeeHistory

This class stores the employment history of staff members.

- **Attributes:**
 - teacherID (int): Unique identifier for the teacher's record.
 - **Operations:**
 - No explicit operations provided in the current context.
-

29. Email_System

This class manages the sending of email notifications within the system.

- **Attributes:**
 - EmailID (int): Unique identifier for the email.
 - EmailAddress (String): The address of the email recipient.
 - **Operations:**
 - sendEmailNotification(notificationMessage: String, EmailAddress: String): void: Sends an email notification to the specified address.
-

30. DataBaseAccess

This class handles database operations, including connecting to the database, retrieving, updating, and deleting data.

- **Attributes:**
 - dbConnection (DatabaseConnection): Represents the database connection instance.
 - data (ArrayList): Stores retrieved or manipulated data.
- **Operations:**
 - addStudentData(Student: student): void: Adds a new student's data.
 - getData(): Student: Retrieves student data from the database.
 - connectDB(): void: Establishes a connection to the database.
 - executeStatement(String: SQL): void: Executes an SQL statement.
 - selectID(ID: int): boolean: Verifies if a specific ID exists in the database.
 - updateName(name: String): void: Updates a student's name.
 - updatePhoneName(phone: String): void: Updates a student's phone number.
 - updateEmail(email: String): void: Updates a student's email address.
 - updateMedicalInfo(medicalInfo: MedicalInfo): void: Updates a student's medical details.
 - updateCourses(courses[]: Course): void: Updates a student's courses.
 - updateAcademicYear(academicYear: int): void: Updates the academic year for a student.

- `updateIsPaid()`: boolean: Updates and checks fee payment status.
 - `deleteInfo(id: int)`: void: Deletes student information by ID.
-

31. Validation

This class validates inputs such as email, ID, passwords, and salaries to ensure data integrity.

- **Attributes:**
 - `MIN_PASSWORD_LENGTH (int)`: Minimum password length required.
 - `EMAIL_REGEX (String)`: Regular expression to validate email formats.
 - `MINIMUM_SALARY (int)`: Minimum allowable salary (default: 0).
 - `ID_REGEX (String)`: Regular expression to validate IDs.
 - **Operations:**
 - `isValidID(id: String)`: boolean: Validates an ID.
 - `isValidEmail(email: String)`: boolean: Checks if an email address is valid.
 - `isValidPhoneNumber(phoneNumber: String)`: boolean: Validates a phone number.
 - `isValidPassword(password: String)`: boolean: Ensures the password meets requirements.
 - `isValidSalary(salary: double)`: boolean: Verifies salary is above the minimum.
 - `isValidDataRange(startDate: Date, endDate: Date)`: boolean: Confirms a valid date range.
-

32. IValidation (Interface)

This interface defines validation methods to be implemented for input verification.

- **Operations:**
 - `isValidID(id: String)`: boolean: Validates an ID.
 - `isValidEmail(email: String)`: boolean: Validates email format.
 - `isValidPhoneNumber(phoneNumber: String)`: boolean: Validates phone numbers.
 - `isValidPassword(password: String)`: boolean: Checks password strength.
 - `isValidSalary(salary: double)`: boolean: Ensures salary is valid.
 - `isValidDataRange(startDate: Date, endDate: Date)`: boolean: Validates date ranges.
-

33. Employee

This class represents employees and their details in the system.

- **Attributes:**
 - `name (String)`: Name of the employee.

- position (String): Job title of the employee.
 - salary (Salary): Salary details for the employee.
 - ID (int): Unique identifier for the employee.
 - email (String): Email address of the employee.
 - phoneNum (long): Phone number of the employee.
 - address (String): Residential address of the employee.
 - dateOfJoining (Date): Date the employee joined the organization.
-

34. Salary

This class manages and calculates salary components for employees.

- **Attributes:**

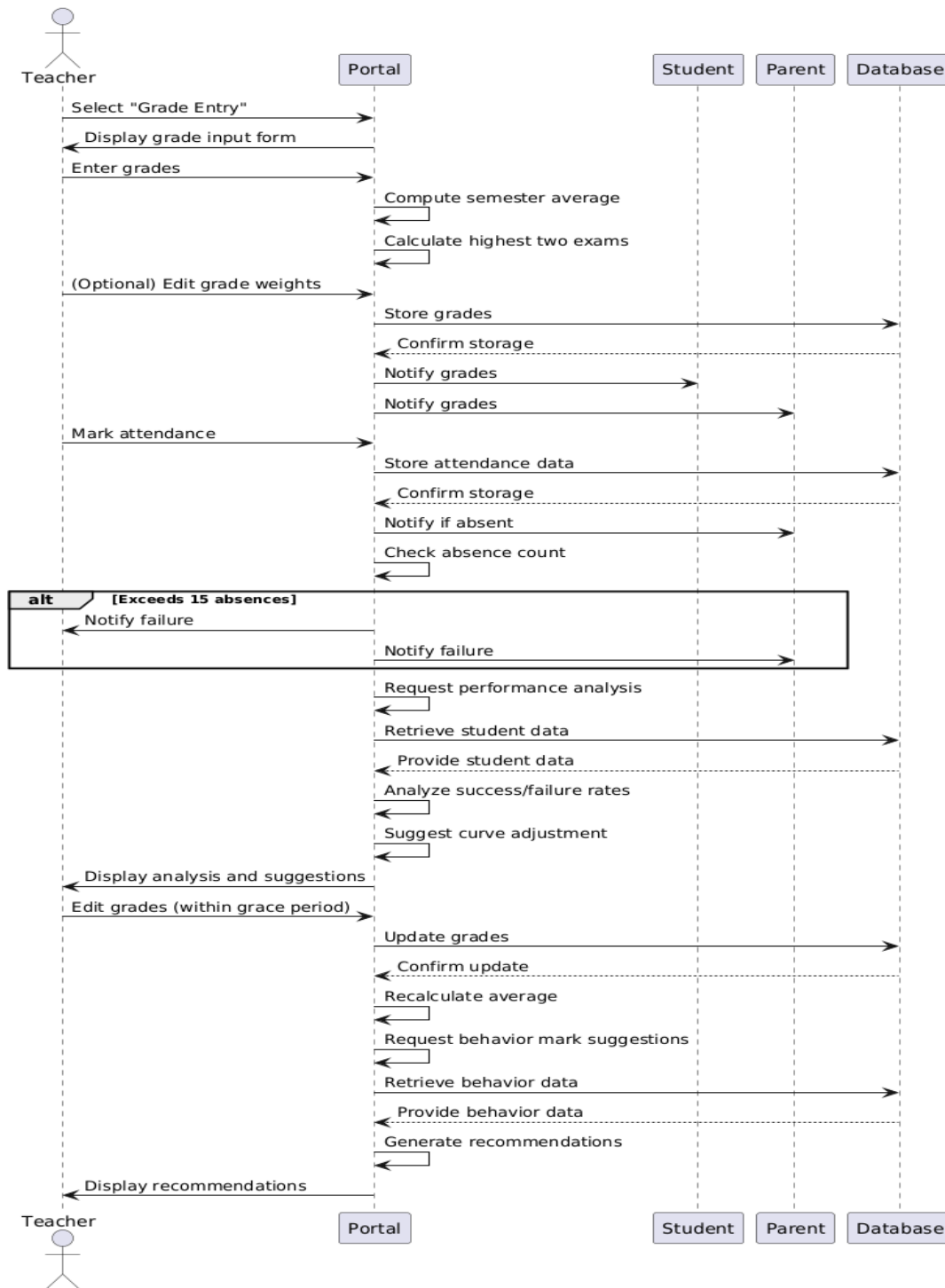
- basicSalary (double): Base salary.
- allowances (double): Additional allowances.
- bonuses (double): Bonuses earned by the employee.
- deductions (double): Salary deductions.
- netSalary (double): Final salary after deductions and additions.
- payPeriod (String): Payment period (e.g., monthly).
- employeeID (int): ID of the associated employee.
- employeeName (String): Name of the employee.
- positionTitle (String): Job title of the employee.
- empType (String): Employment type (e.g., full-time, part-time).
- Emp (Employee): Reference to the associated Employee object.

- **Operations:**

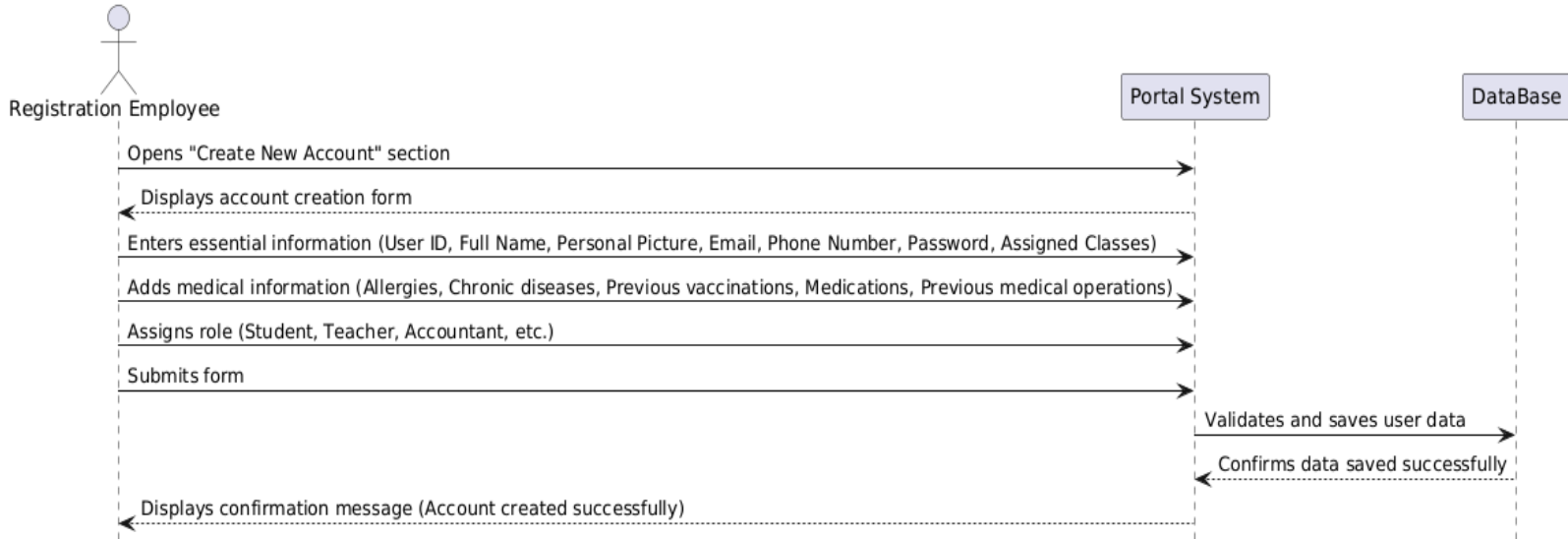
- getBasicSalary(): double: Retrieves the basic salary.
- getAllowances(): double: Retrieves allowances.
- getBonuses(): double: Retrieves bonuses.
- getDeductions(): double: Retrieves deductions.
- getNetSalary(): double: Retrieves the net salary.
- getPayPeriod(): String: Retrieves the payment period.
- getEmployeeID(): int: Retrieves the employee ID.
- getEmployeeName(): String: Retrieves the employee's name.
- getPositionTitle(): String: Retrieves the employee's job title.
- calculateNetSalary(): double: Calculates the net salary.
- updateBasicSalary(newBasicSalary: double): void: Updates the basic salary.
- displaySalaryDetails(): void: Displays salary details.
- resetSalary(): void: Resets salary components to default values.

Task 4.2: [Individual Task] System Sequence modelling and Analysis:

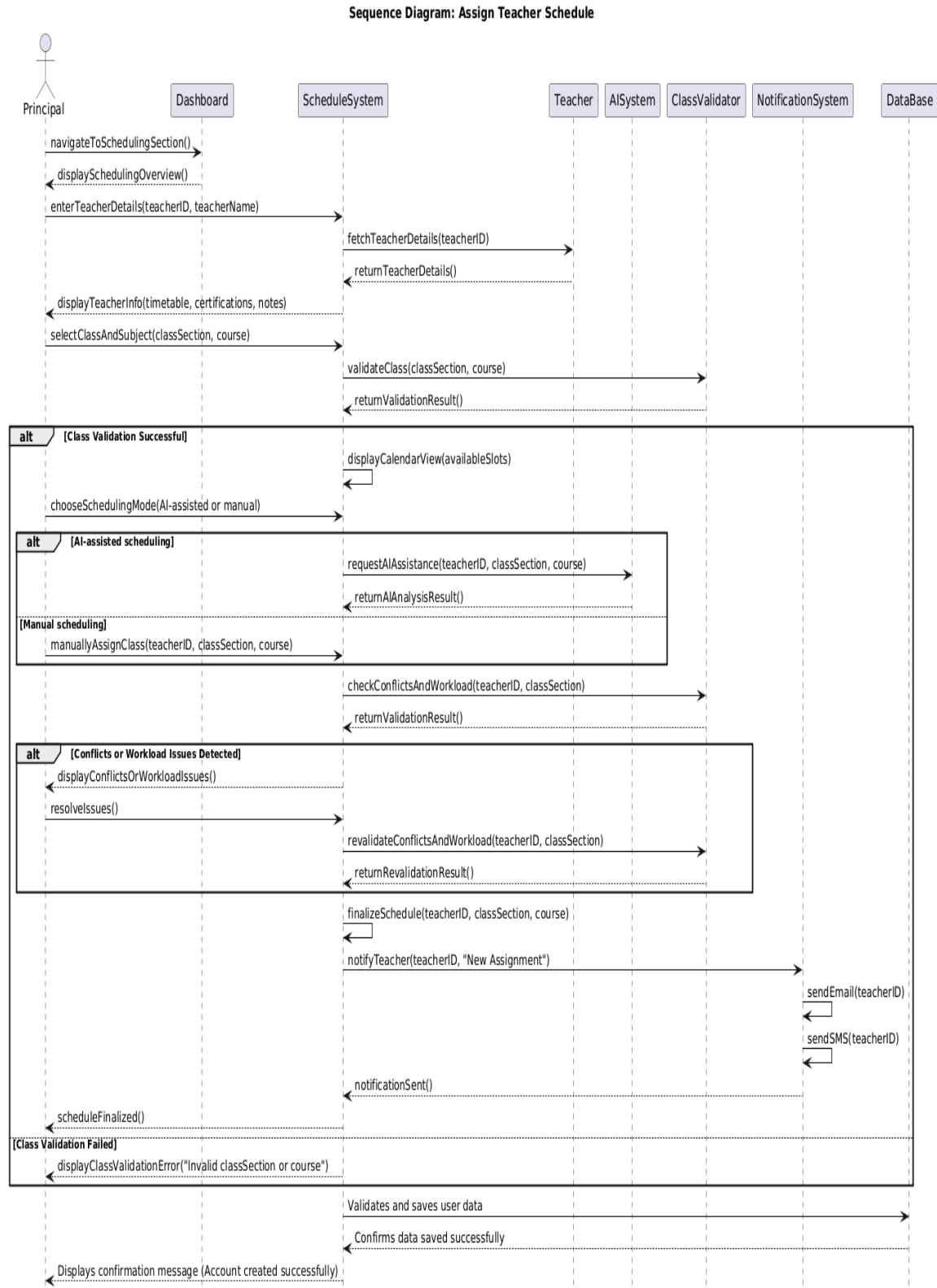
Publishing Grades Sequence Diagram (Rand Amly):



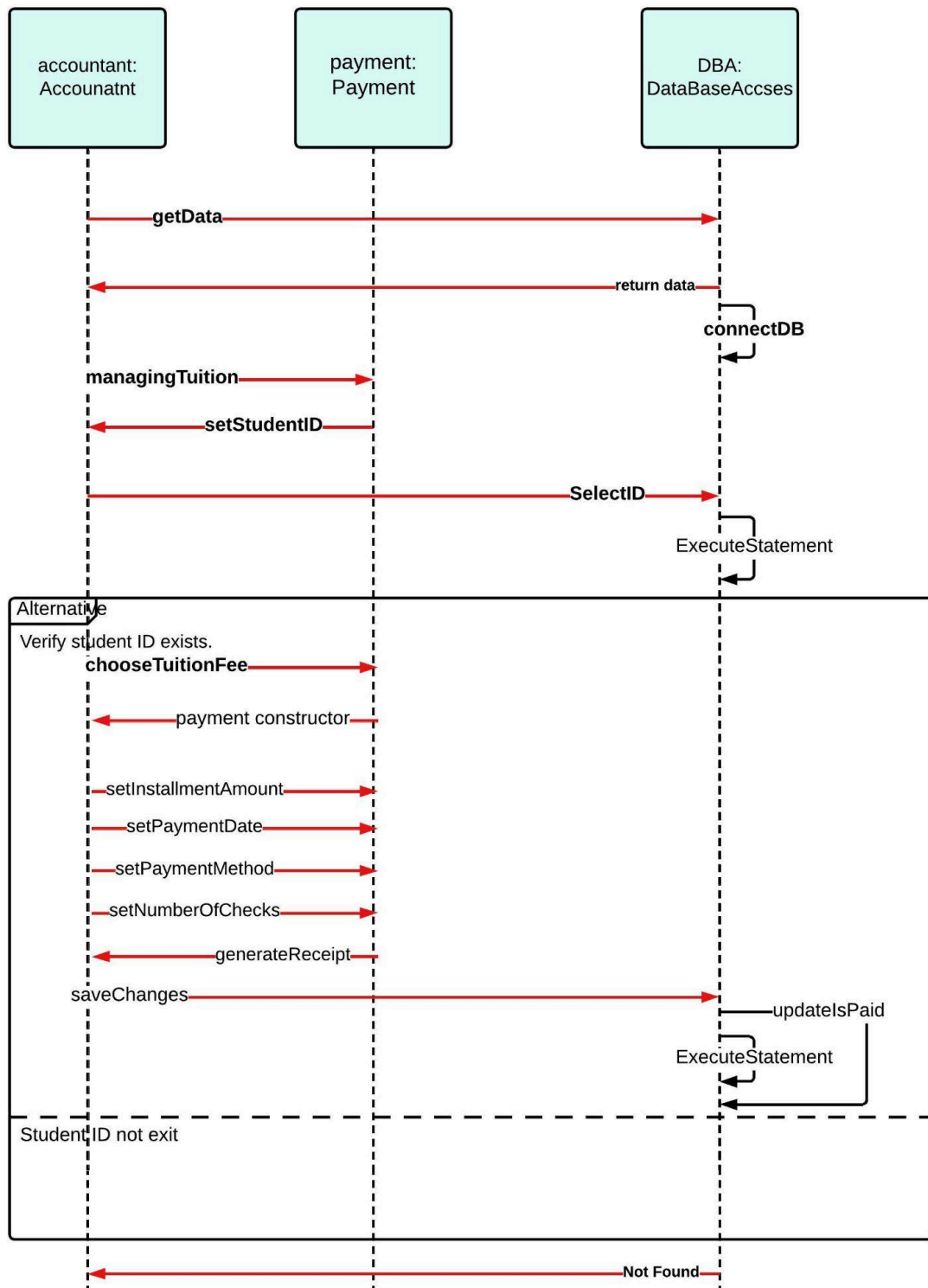
Creating a new account and assign role for a new user Sequence diagram (Wasfia Awwad):



Assign Teacher Schedule Sequence diagram (Sadeel Jabareen):



Student payment management Sequence diagram (Toqa Abdeen) :



Task 4.3: [Group Task] System Design Goals:

1. General design goals:

a. Low Coupling:

goal: The dependency between different system modules or parts is minimum and as low as possible.

Rationale: With loose coupling, changes in one module would have little impact on other modules; thus, making the system easier to maintain, test, and scale. Quite important for a system such as AcademiaHub, since many of the departments dealing with one another such as registrations, accounting, and management.

How to Implement:

Modules will communicate via well-defined interfaces.

System design should be modular; each module should have one single responsibility.

Not allow modules to depend directly on other modules; instead, drive in a layer of indirection such as the service layer or API.

b. High Cohesion:

goal: Every module or component shall deal with only one well-defined responsibility.

Rationale: High cohesion has the effect of making the code more readable, maintainable and reusable. Each module is self-contained with some form of role, which guarantees that changes cannot likely involve unexpected side effects.

How to Implement:

Group similar functionalities into one module. For instance, everything about payment should be in the accounting module. Do not put different functionalities into one module. Mention the usage of design patterns such as SRP so that each class or module will be responsible for something.

2. Specific System Design Goal (Derived from Non-Functional Requirements):

a. Performance:

Architecture Goal:

Response time for requests coming from the students shall not be more than the threshold, that tends to be very high those few days in which final grades are published and deadlines are near for fees settlement.

Performance can be considered one of the critical concerns of user satisfaction since the target application will involve a school management system where timely access to some information-like grades, schedules, and payment status, among others-are crucial. A slow response to such queries reflects inefficiency and frustration.

Implementation Approach:

- **Optimize Database Queries:** Ensure database queries perform faster by making use of indexing and caching where applicable.
- **Load Balancing:** Do the load balancing in distributing the requests of the users to avoid congestion over one server.
- **Caching:** Caching retrieves the information that has been accessed often, such as class timetables of students or grade reports, from available memory rather than querying the database for the same data again and again.

- **Asynchronous Processing:** When immediate feedback is not necessary-for example, in the sending of notifications-use asynchronous processing to further improve the response times of the primary operations.
- **Scalability:** The system shall be designed to scale horizontally. That is, an increase in the number of users is handled by adding servers.

b. Usability:

Architecture Goal:

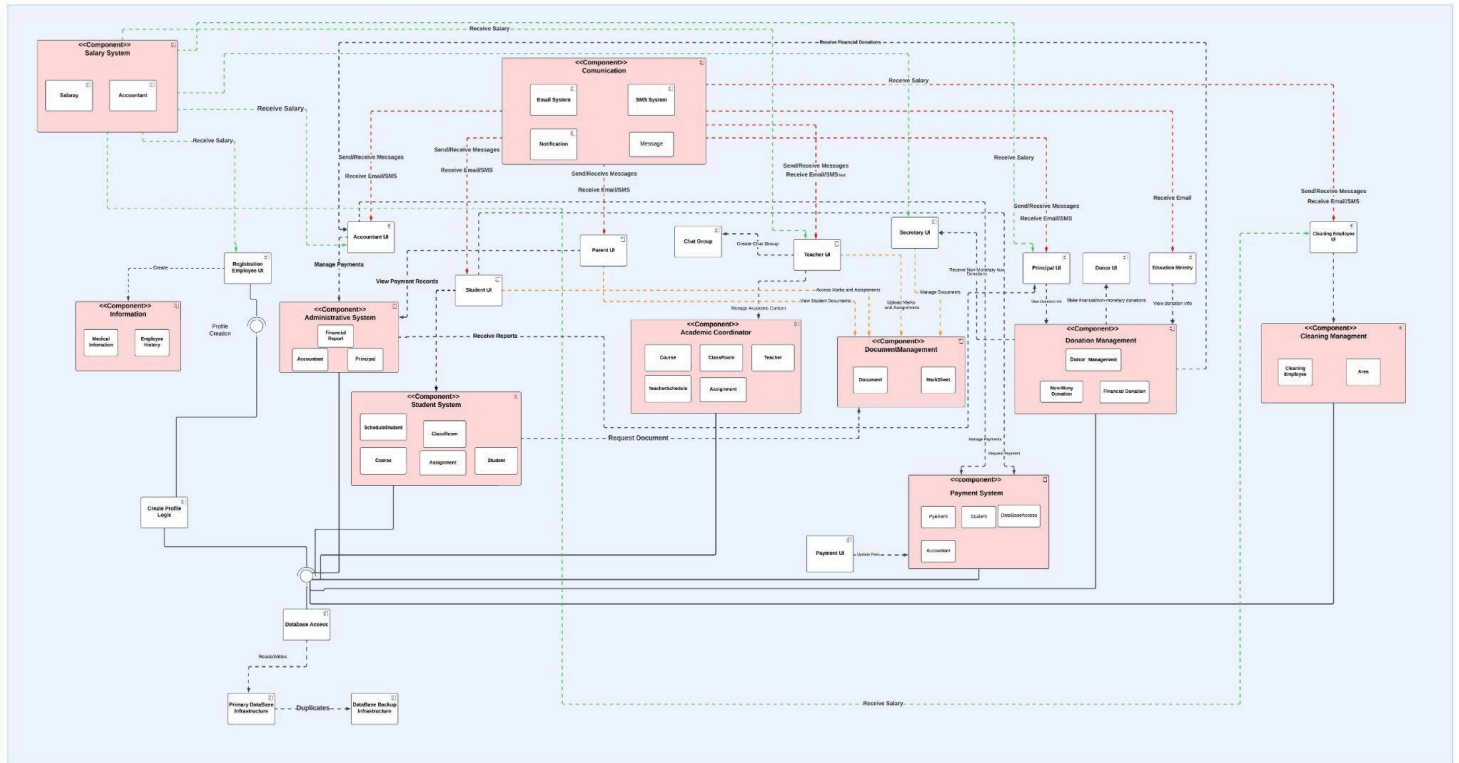
To create an accessible and user-friendly school portal system that accommodates the needs of school staff, students, parents, and students with special needs. The portal will provide seamless navigation, intuitive features, and accessibility options like text-to-audio conversion and subtitles for audio content. Additionally, it will offer interactive elements such as quiz games and personalized guidance for students in choosing academic fields based on their performance.

Implementation Approach:

- **Inclusive Design:** The portal will incorporate accessibility features such as text-to-audio conversion and subtitles for audio content, ensuring that students with special needs can fully engage with the system. These features will be designed to comply with accessibility standards, enhancing the portal's usability for all users.
- **User-Friendly Interface:** A clear and intuitive interface will be developed, catering to a diverse user base, including school staff, students, and parents. The design will prioritize ease of use, with simple navigation and visually distinct sections for different user roles.
- **Interactive Learning Tools:** The portal will include interactive features like quiz games to support student engagement and learning. Additionally, a personalized guidance system will help students explore potential academic fields based on their grades and interests, making the portal a valuable resource for academic planning.
- **Comprehensive Onboarding and Support:** To ensure smooth adoption, the portal will offer an initial guidance module for new users, providing step-by-step

instructions on how to navigate and utilize key features. Continuous feedback mechanisms will be in place, allowing for iterative improvements and user-driven enhancements.

Task 4.4: [Group Task] System Component Design:



Task 4.5: [Group Task] System Deployment Design:

